

Chapter 1

Introduction

1.1 Waves in the Ocean and Their Significance

Atmosphere and ocean form a coupled system which continuously exchanges heat, momentum and mass at the air-sea interface governing the dynamics of weather, climate and global change. Due to energy flow from the atmosphere to the ocean this interface presents an aerodynamically rough surface that can comprise dynamic, unsteady, very high and steep surface waves, which are the results of the various external forces of nature dictating what types of waves can be induced in the ocean. A realistic description of the physical processes occurring at the ocean-atmosphere interface is essential for a reliable determination of the air-sea fluxes of momentum, sensible and latent heat, CO_2 and other trace gases, and aerosols. As the wave field is involved in these exchange processes, wave models should be able to compute not only the wave spectra and wave statistics, but also the processes at the air-sea interface that govern the fluxes across the interface.

The most obvious cause of surface waves is an action of wind. The ancient Greeks were well aware of interaction between the atmosphere and sea surface. Aristotle (384–322 BC) realised that wind acting on the sea surface plays a very important role in the development of waves. Pliny (AD 23–79) observed that oil poured upon waves calms them. From the time of Aristotle to the Renaissance of science in the ‘Golden Age of Discovery’ in the late fifteenth century, very little progress was made towards an understanding of the generation and growth of waves. It was not until well into the nineteenth and twentieth centuries that more fundamental knowledge of what causes waves and how they behave was accumulated.

At present, the study of surface wave mechanics are concentrated predominantly on three aspects. On the one hand, wave mechanics was

treated as a pure mathematical problem, important also for numerical modelling (Stokes, 1847, 1880; Lamb, 1932; Stoker, 1957; Phillips, 1966, 1977; LeBlond and Mysak, 1978; Davidan *et al.*, 1978; Mei, 1983; Komen *et al.*, 1994; Massel, 1989, 2007, 2013; Dingemans, 1997; Ochi, 1998; Kantha and Clayson, 2000; Lavrenov, 2003; Pedlosky, 2003; Janssen, 2004; Mei *et al.*, 2005; Holthuijsen, 2007; Babanin, 2011 and others).

On the other, dynamics of surface waves were used to estimate the resultant wave loads on offshore and coastal structures under extreme and operating conditions. Textbooks and scientific papers provides many examples of the direct applications of wave mechanics to ocean and coastal engineering (Krylov *et al.*, 1976, 1986; Sarpkaya and Isaacson, 1981; Dean and Dalrymple, 1998; Goda, 1985, 2000; Mei *et al.*, 2005 and others).

In recent years, an increasing attention has been directed to climate change and the role played by the oceans. The interaction between the ocean and atmosphere at the air-sea interface is critical to our understanding of the earth's climate as ocean surface forms a filter to the exchange of heat, moisture, momentum and trace constituents. The energy transfer from the atmosphere to the ocean enhances heat flux and mixed layer during the circulation of the upper ocean, while the energy fluxes from the ocean to the atmosphere affect gas and aerosol concentration, as well as the atmospheric circulation, weather and climate. Oceans are not empty, but support a large variety of organisms and plants. Ocean waters interact, influence and constrain the ocean's biological life. In particular, ocean waters are a critical factor for enhancement of mixing in the upper ocean layer and transferring of oxygen and nutrients into water column (Mann and Lazier, 1996; Massel, 1999, 2007; Janssen, 2004; Babanin, 2011).

In general, five basic types of waves, present in the ocean can be distinguished: sound, capillary, gravity, internal and planetary waves. Sound waves are due to water compressibility, which is, in fact, very small. Gravity forces, acting on water particles displaced from equilibrium at the ocean surface or at an internal geopotential surface in a stratified fluid, induce gravity waves (surface or internal). The combination of the turbulent wind and surface tension gives rise to short, high frequency capillary waves. On the other hand, very slow, large-scale planetary or Rossby waves are induced by the variation of the equilibrium potential vorticity due to changes in depth or latitude. All of the above wave types can occur together, producing more complicated patterns of oscillations.

Moreover, the gravitational attraction between the Earth, Moon and Sun generate tides, which are very long oceanic waves with periods of the

order of 12 hours and more. Complexity of land contours on the Earth and complicated bathymetry cause great difficulties in the prediction of tides at some points on the Earth. However, the tides can still be predicted using the harmonic analysis of a sufficiently long record of water level fluctuation.

Under specific conditions, long waves can also be generated in closed water basins such as lakes, bays, estuaries and harbours, and they are known as seiches. The most probable sources of seiches forcing mechanisms are meteorological disturbances, impact of wind gusts on the water surface, seismic oscillations of the earth during earthquakes and others. The frequency of the free oscillations (natural oscillations) are determined by the shape of the basin and water depth. However, the meteorological conditions may considerably change the height of predicted seiches and complicate the prediction process.

The oscillations of water level known as the storm surges, are the disturbances of sea level due to meteorological factors resulting from wind, atmospheric pressure changes, moving storm systems and bottom configuration. Storm surges may be expected to occur along coasts with relatively shallow waters which are affected by the passage of storms.

Strong earthquakes with an epicentre in or near an ocean basin generate large disturbances of the ocean surface known as tsunami. Tsunami are impulsively driven water waves caused by the sudden displacement of a large mass of water. They propagate radially from their origin on the ocean surface and their impact is particularly severe on the ocean coasts, ranging from erosion and over-washing of beaches, localised or regional floods to the complete devastation of coastal villages and structures. Occasionally, during special wave events appearing in nature, such as meteorites impact on ocean surface or glaciers calving in polar regions, impulsive type waves of the multi-frequency nature are generated.

In general, the frequency range associated with external forces is very wide and ocean surface response occupies an extraordinarily broad range of wave lengths and periods, from capillary waves, with periods of less than a second, through wind-induced waves and swell with periods of the order of a few seconds, to tidal oscillations with periods of the order of several hours and storm surges with periods of the order of days. In Fig. 1.1 and in Table 1.1, the schematic representation of energy contained in the surface waves, and the physical mechanisms generating these waves, are listed. The Figure gives some impression of the relative importance of various kinds of surface oscillations, but does not necessarily reflect the actual energy content.

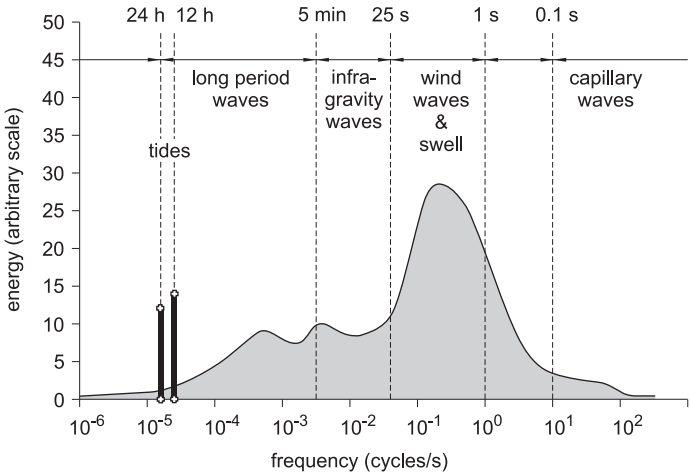


Fig. 1.1 Schematic distribution of wave energy in frequencies.

Table 1.1 Waves, physical mechanisms, and periods.

Wave type	Physical mechanism	Periods
Capillary waves	Surface tension	$< 10^{-1}$ s
Wind waves	Wind shear, gravity	< 15 s
Rogue waves	Nonlinear mechanisms	< 15 s
Swell	Wind waves	< 30 s
Surf beat	Wave groups	1–5 min
Seiche	Meteorological disturbances	2–40 min
Internal waves	Density structure instability	10 min – 15 h
Tsunami	Earthquake	10 min – 2 h
Tides	Gravitational action of the moon and sun, earth rotation	12–24 h
Storm surges	Wind stresses and atmospheric pressure variation	1–3 days

For offshore engineering operations, offshore structures and ship performance, gravity waves are of the greatest importance, as the influence of the loads due to gravity waves on engineering structures is the most critical and hostile. Marine structures must be designed to sustain the forces and velocities induced by waves. An accurate assessment of the maximum water particle velocities beneath the high wave crest is required for drag

force calculations, while the maximum accelerations beneath the steepest section of the wave profile are critical for inertial force calculations. Therefore, a thorough understanding of the interaction of waves with offshore structures has now become a vital factor in the safe and economical design of such structures. The calculation procedures necessary to establish the structural loading generally involve the few steps in which knowledge on the surface waves is essential, namely establishing the wave climate in the vicinity of the structure, estimating design wave conditions for the structure, and selecting and applying a wave theory to determine the hydrodynamical loading on the structure. In particular, the exceptionally large waves, called rogue waves or freak waves, represent a danger to ships and marine structures. A number of ship and offshore platform accidents related to the action of rogue waves have been reported in the literature.

The role of waves in coastal marine environment can not be overestimated. Waves approaching the shoreline break and dissipate their energy on beaches. Storm and cyclone waves impose large forces on natural coastal and man made structures. Long-shore currents transport sediments and create areas of erosion and accumulation. The waves are very important for transporting energy, sediment and chemical and biological matter. Knowledge of wave motion and the sediment budget provides the key to the proper selection of protecting structures and methods of shoreline stabilisation. Our present understanding of these processes is still not adequate to develop effective management plans or engineering designs which are today subjected to very stringent requirements to minimise their impact on the environment. Being dependent on the quality of the environment, human activities are recognised as drivers of significant pressures on marine and coastal biodiversity, pollution, and nutrient emissions.

Other types of ocean waves which play a smaller role in ocean and coastal engineering or have been treated in other books are not included here. In particular, for a systematic review of planetary waves, tides and storm surges, internal waves and coastal trapped Kelvin waves I refer the reader to the books by Phillips (1977), LeBlond and Mysak (1978), Pedlosky (1986), Monin (1988), Cushman-Roisin (1994), Miropolsky (2001), Dijkstra (2005), Flór (2010), Sutherland (2010) and Massel (2015).

1.2 Basic Assumptions on Seawater and Wave Motion

1.2.1 Continuous fluid and water particle concepts

Prior to introducing the basic concepts on fluids and fluid motion, we will adopt the rectangular coordinate system $O(x, y, z)$ or $O(x_i, z)$; $i = 1, 2$. The origin of the system is at the mean sea surface. The axes x and y are horizontal. The z axis is directed opposite to the force of gravity.

A fluid is composed of a large number of molecules which are in constant motion and underlying collisions with each other. Therefore when studying the mechanics of a fluid it is possible in principle to consider the motion of the fluid molecules themselves, as it done in kinetic theory or statistical mechanics. However, the average quantities are usually sufficient for the explanation of the observed macroscopic phenomena. Therefore, the discrete molecular structure is replaced with a continuous distribution, known as *continuum*. By continuous fluid in motion, we mean that the velocity $\mathbf{u}$ is everywhere finite and continuous while its space derivatives of the first order are finite (but not necessarily continuous). Thus, any closed surface S which moves with the fluid, permanently and completely separates the fluid matter inside S from that outside. The fluid can be treated as continuous at the scale A (obstacle dimension, a pore diameter, a turning radius et.) when the flow past an obstacle of the dimension A is much larger than the average free path of the molecule l_0 (for water $l_0 \approx 3 \cdot 10^{-10}$ m). The mean free path specifies the average distance that molecule travels before it demonstrates its presence, temperature, or momentum to other molecules. The rate (l_0/A) is known as the Knudsen number (Kn). If

$$Kn = \frac{l_0}{A} < 0.01, \quad (1.1)$$

the fluid can be treated as continuous (Kundu *et al.*, 2016). For most terrestrial situations, above requirement is not a great restriction. However, for flow over small objects or in very narrow passages, the molecular-kinetic approach is necessary.

With regard to continuous fluid, we can define a *fluid particle* as consisting of the fluid contained within an infinitesimal volume, that is to say, a volume whose size may be considered so small that for the particular purpose in hand its linear dimensions are negligible. We can then treat a fluid particle as a geometrical point. According to continuum hypothesis, fluid properties like temperature, density, or velocity are defined at every point in space.

1.2.2 Properties of seawater and its motion

The equations of motion for the fluid particle depends on the physical properties of the fluid and motion itself. In order to render the subject amenable to exact mathematical treatment, we examine some physical and chemical properties of seawater and simplifying assumptions on the medium and its motion, i.e.:

(a) seawater temperature, salinity and density

Seawater is not a pure water. It contains salts, dissolved organic substances, and mineral and organic suspended matter. Water has many unique chemical and physical properties. Unlike most substances which contract when frozen, water expands, allowing ice to float on the surface of water basins. The principal seawater state parameter is *density*, or mass per unit volume. The density of water controls many fundamental processes in the ocean, e.g. the hydrostatic equilibrium or motion of water particles and propagation of internal waves, turbulence and mixing in water column, heat transfer, concentration of plankton and sediment transport, locomotion of marine species and many others. Even small alterations in the density of seawater result in great changes in water flow and its thermal and chemical status.

Density of water is normally symbolised by ρ_w and in the SI system it is expressed in kilogram per cubic meter (kg m^{-3}). For example, the density of pure water is approximately 1000 kg m^{-3} being 770 to 890 times that of air, ρ_a , at sea level. Unlike measuring salinity, temperature and pressure, there are no practical methods of measuring the density of seawater *in situ*. The density of seawater is usually determined through its dependence on pressure, p , temperature, T , and salinity, S . Pressure, p , has an insignificant effect on the density of water for most applications, unless one is dealing with water at great depths within the ocean.¹ The density is much more dependent on *temperature*, generally decreasing as the temperature increases. However, the density of freshwater is not a monotonic function of temperature, and water density reaches a maximum at temperature of 3.98°C under normal atmospheric pressure.² This anomalous dependence is a result of the specific molecular structure of water (Dera, 1992).

¹For a depth of 1 km the density of water is only 0.5% greater than at the surface, and at the deepest point in the earth's oceans, the Marianas Trench, water is about 6% denser than water at the surface, assuming that the temperature and salinity are constant (Denny, 1993).

²Normal atmospheric pressure at sea level, called the *normal atmosphere*, is defined to be $1.01325 \cdot 10^5 \text{ Pa}$ at 15°C ($1 \text{ Pa} = 1 \text{ N m}^{-2}$).

The mean temperature, T , of the surface water of oceans is about 17.8°C and it is 3.6°C higher than the mean air temperature on the Earth's surface. In the Pacific Ocean, the mean water temperature is equal to 19.37°C , in the Atlantic Ocean – to 17.58°C and in the Indian Ocean – to 17.85°C , while at the Arctic Ocean depth it is about $(1.0\text{--}1.5)^{\circ}\text{C}$.

The relatively constant composition of the main constituents of sea salt has made it possible to introduce a single parameter defining the salt concentration in seawater. This is known as *salinity* (S). Away from coasts, the salinity varies from 32 to 37 PSU (Practical Salinity Units) and the mean salinity of the ocean waters (excluding Arctic waters) is equal to 34.71 PSU. There is a weak correlation between salinity and the latitude. In general, cold waters are less saline than warmer waters. In closed or semi-closed seas, such as the Baltic Sea, salinity is equal to 7–8 PSU, being, however, dependent on the rivers discharge, and the difference between precipitation and evaporation.

Practical Salinity, S , is today a globally accepted standard for reporting the salinity of seawater and has been in use since 1978. The Practical Salinity Scale, PSS-78, is based on a carefully determined salinity-conductivity relationship. According to the formulation of PSS-78, the Practical Salinity, S , of a sample of seawater, is defined in terms of the ratio K_{15} of the electrical conductivity of seawater at the temperature of 15°C and the pressure of one standard atmosphere, to that of a potassium chloride (KCl) solution, in which the mass fraction is $32.4356 \cdot 10^{-3}$, at the same temperature and pressure. A K_{15} value exactly equal to 1 corresponds, by definition, to a Practical Salinity exactly equal to 35 (UNESCO, 1981).

Except the Practical Salinity Scale, the so called Absolute Salinity S_A of seawater has come under discussion. The Absolute Salinity S_A is defined as the ratio of mass of dissolved material in seawater to the total mass of seawater. In principle, the dissolved material includes not only those compounds in water conducting electricity. Such components are usually present in many local sea basins. However, the S_A quantity cannot be measured directly. Millero *et al.* (2008) proposed a new salinity variable, called the Reference-Composition Salinity, or Reference Salinity, to provide a means to more accurately estimate the Absolute Salinity S_A of a local seawater sample.

Density of seawater $\rho_w(S, T, p)$ for temperature T ($0 < T < 40^{\circ}\text{C}$), salinity S ($0 < S < 42$ PSU) and pressure p can be determined from the

International Equation of State for Seawater (UNESCO, 1981; Dera, 1992)

$$\varrho_w = \frac{\varrho_w(S, T, 0)}{1 - \frac{p}{K(S, T, p)}}, \quad (1.2)$$

where $K(T, S, p)$ is module of seawater compressibility. The expression for $\varrho_w(S, T, p)$ is rather complicated and it will not be given here.³ For illustration, some values of density $\varrho_w(S, T, 0)$ are listed below:

$$\begin{aligned} \varrho_w(0, 10^\circ, 0) &= 999.702 \text{ kg m}^{-3}; \\ \varrho_w(35, 10^\circ, 0) &= 1026.952 \text{ kg m}^{-3}; \\ \varrho_w(0, 20^\circ, 0) &= 998.206 \text{ kg m}^{-3}; \\ \varrho_w(35, 20^\circ, 0) &= 1024.763 \text{ kg m}^{-3}. \end{aligned}$$

In place of the water density, the specific weight will be sometimes used in this book. This quantity, usually symbolised by γ , is equal to the product of water density ϱ_w and gravitational acceleration, g , so $\gamma = \varrho_w g$. The specific weight is expressed in newtons per cubic metre (N m^{-3}).

The density of seawater and its distribution in the water column determines the hydrostatic stability of water masses and influences the turbulence and internal waves and sound propagation. However, the influence of density on surface waves is negligible, except perhaps for the stage of wave generation under wind action (see Chapter 2).

(b) *water is an inviscid and incompressible fluid*

Viscosity is a property which is a measure of a fluid's resistance to 'deformation' during motion. Within a fluid, momentum of rapidly moving particle is exchanged with the momentum of relatively slower particles. Those exchanges produce a shearing stress. In general we can write for shear stress τ

$$\tau = \mu \frac{d\mathbf{u}}{d\mathbf{n}}, \quad (1.3)$$

where τ is tangential stress, $\mathbf{u} = (u, v, w)$ is fluid velocity vector, $\mathbf{n}$ denotes vector normal to vector $\mathbf{u}$, and μ is the *coefficient of dynamic molecular viscosity* of the fluid. It has units of (Ns m^{-2}). A fluid which shows the direct proportionality between the applied shear stress and the resulting rate of deformation is called a *Newtonian fluid*. Many biological materials, such as blood, synovial fluid, mucus of various consistencies, can not be treated as Newtonian fluids. Some of them have a memory of previous

³The algorithm of determination of density for given salinity, S , temperature, T , and pressure, p , according to UNESCO Formula is given in UNESCO, 1981 or Massel, 2015.

shape and elasticity. A description of such fluids is a subject of another branch of fluid mechanics, so-called rheology.

As force on an element of fluid varies like τ but the mass of that element varies like ϱ_w , the acceleration and hence the velocity field is determined by the ratio τ/ϱ_w , known as the *coefficient of kinematic viscosity*, ν

$$\nu = \frac{\mu}{\varrho_w}, \quad (1.4)$$

where ν has units of $\text{m}^2 \text{s}^{-1}$. For example, the coefficient of kinematic viscosity for seawater of salinity $S = 35$ PSU and of temperature $T = 20^\circ\text{C}$ becomes $1.064 \cdot 10^{-6} \text{m}^2 \text{s}^{-1}$. It is interesting to note that there is almost no liquid with viscosity lower than that of water, and that air viscosity $\nu_{\text{air}} \approx 15 \nu_{\text{water}}$.

Both coefficients, μ and ν , are physical properties of fluid, independent of fluid motion. When a fluid is completely at rest or is moving very slowly, the rate of diffusion of momentum is essentially determined by molecular motion. Eddies are small and the velocity shear is great, and the internal resistance of the water due to molecular viscosity smooths out the gradients in velocity. This smoothing of the flow is the way the energy in turbulence is finally converted to heat and dissipated.

However, as soon as the fluid is stirred and the elements of fluid are much greater than molecules mix, the coefficient μ increases considerably. In the oceans and atmosphere, eddies and turbulent motions in the flow can be so effective in moving particles among themselves that the effects of molecular diffusion are overwhelmed. For such situations, the coefficient of kinematic viscosity, ν , is replaced by the *coefficient of turbulent viscosity*, which is several hundred to many thousand times larger. By analogy to molecular exchange, the turbulent viscosity coefficients result from the hypothesis that the turbulent momentum flux is proportional to the averaged turbulent flow velocity. Here we only note that the estimates of the coefficient of turbulent viscosity in the ocean vary enormously, from $10 \text{m}^2 \text{s}^{-1}$ to $10^4 \text{m}^2 \text{s}^{-1}$ in the horizontal plane, and from $10^{-4} \text{m}^2 \text{s}^{-1}$ to $10^{-1} \text{m}^2 \text{s}^{-1}$ in the vertical plane.

The ratio $Re = uL/\nu$ (the Reynolds number), in which L is the differential length scale and u is the flow velocity, represents the relative magnitudes of the inertial and viscous terms. In many oceanic motions, the Reynolds number is very large. Thus, the viscous influence is often quite negligible over most of the field of motion. The viscous forces are important only in narrow regions of the flow, where the local inertial and viscous forces are comparable. In the ocean, the interfacial layer between the air and the water, as well as the bottom boundary layer are such regions. The

thickness δ of the surface boundary layer is of the order $\delta \approx (2\nu/\omega)^{1/2}$, where ω is a wave frequency, and ν is kinematic coefficient of viscosity. For typical ocean wave frequencies, the thickness $\delta \sim 0.001$ m. For the boundary layer near the natural sea bottom, the eddy viscosity is much higher. The thickness of the boundary layer is then about 0.1 m, which is still quite small. Therefore, the boundary layer regions are but a very small fraction of a fluid volume, and the influence of the molecular viscosity on the wave motion can be neglected.

The *compressibility* of water is very small and the Young's modulus is of order $E \approx 3.05 \cdot 10^8 \text{ N m}^{-2}$ (Dera, 1992). As the typical velocity of seawater is much smaller than the speed of sound, the very small water compressibility has no influence on water motion.

(c) conservation laws

The basic principles in fluid mechanics are the conservation laws for mass, momentum and energy. In the following we present them in differential form, applicable at a point or to a fluid particle. Therefore, conservation of mass takes the form of equation

$$\frac{\partial}{\partial t}(\varrho_w) + \frac{\partial}{\partial x}(\varrho_w u) + \frac{\partial}{\partial y}(\varrho_w v) + \frac{\partial}{\partial z}(\varrho_w w) = 0. \quad (1.5)$$

For constant density throughout the flow field, and for incompressible flow from Eq. (1.5) we have

$$\text{div } \mathbf{u} = \nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (1.6)$$

where

$$\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}. \quad (1.7)$$

The equations of motion for fluid provide the relationship between the applied forces on a unit volume of fluid and the change of momentum. The basic equation for derivation of the equation of motion is Newton's second law.

For incompressible fluid the equation of motion are called the Euler equations⁴

$$\left. \begin{aligned} \varrho_w \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) &= -\frac{\partial p}{\partial x} \\ \varrho_w \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) &= -\frac{\partial p}{\partial y} \\ \varrho_w \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) &= -\frac{\partial(p + \varrho_w g z)}{\partial z} \end{aligned} \right\}, \quad (1.8)$$

or in vector notation

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \left(\frac{1}{2} |\mathbf{u}|^2 + g z + \frac{p}{\varrho_w} \right) + (\text{curl} \mathbf{u}) \times \mathbf{u} = 0, \quad (1.9)$$

in which

$$\begin{aligned} \text{rot} \mathbf{u} &= \text{curl} \mathbf{u} = \nabla \times \mathbf{u} \\ &= \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \mathbf{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{k}, \end{aligned} \quad (1.10)$$

where u , v and w are the velocity components on x , y and z axes, respectively. The vector $\text{rot} \mathbf{u}$ is known as the vorticity. When the vorticity is different from zero, the motion is defined as *rotational*.

Some flows in the ocean are rotational due to presence of viscosity friction and the Earth's rotation; they also reveal a lot of diffusion. Extension of the Euler equations to viscous fluids is given in the form of the Navier-Stokes equations

$$\left. \begin{aligned} \varrho_w \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) &= -\frac{\partial p}{\partial x} + \mu \nabla^2 u \\ \varrho_w \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) &= -\frac{\partial p}{\partial y} + \mu \nabla^2 v \\ \varrho_w \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) &= -\frac{\partial(p + \varrho_w g z)}{\partial z} + \mu \nabla^2 w \end{aligned} \right\}. \quad (1.11)$$

In many oceanic motions the influence of the viscous terms is quite negligible. In such case, the Lagrange theorem indicates that if, at some initial instant, the vorticity vanishes everywhere in the field of flow, the motion is irrotational. Therefore from Eq. (1.10) is $\text{rot} \mathbf{u} = 0$ or

$$\frac{\partial w}{\partial y} = \frac{\partial v}{\partial z}, \quad \frac{\partial u}{\partial z} = \frac{\partial w}{\partial x}, \quad \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}. \quad (1.12)$$

⁴Leonhard Euler (1707–1783) was a Swiss mathematician and physicist who made very important discoveries in many branches of mathematics, mechanics and fluid dynamics.

The above conditions can be satisfied when the velocity $\mathbf{u}$ is represented as the gradient of a scalar function, the velocity potential Φ

$$\mathbf{u} = \nabla \Phi. \quad (1.13)$$

Then, in virtue of the continuity equation (1.6), the potential Φ obeys the Laplace equation

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0. \quad (1.14)$$

It should be noted that velocity potential $\Phi(x, z)$ and stream function $\psi(x, z)$ constitute in a vertical plane (x, z) the analytical function known as complex velocity potential $w(x, z) = \Phi(x, z) + i\psi(x, z)$, satisfying the Cauchy–Riemann relations

$$\frac{\partial \Phi(x, z)}{\partial x} = \frac{\partial \psi(x, z)}{\partial z}, \quad \frac{\partial \Phi(x, z)}{\partial z} = -\frac{\partial \psi(x, z)}{\partial x}. \quad (1.15)$$

1.3 Fundamentals of Description of Regular Waves

1.3.1 Linear description of waves

The deterministic description is a natural starting point for the development of random wave models. Thus, let us consider a wave train observed at some point $P(x, y)$. The surface displacement of a wave travelling at an angle θ to the x axis may be presented as

$$\zeta(x, y, t) = a \cos [k(x \cos \theta + y \sin \theta) - \omega t + \varphi], \quad (1.16)$$

in which a is the wave amplitude, φ is a phase shift and k is a wavenumber related to wave frequency ω by the classical dispersion relation

$$\omega^2 = gk \tanh(kh). \quad (1.17)$$

Solution of Eq. (1.17) in terms of the wavenumber k is given by Eq. (3.111).

The simplest and the most natural representation of the confused sea surface would be the linear superposition of many harmonics travelling in various directions. A simple illustration of such a superposition is given in Fig. 1.2, where 13 elementary components sum to form a final wave surface displacement. Thus, using Eq. (1.16), the surface displacement may be written as

$$\zeta(x, y, t) = \sum_{n=1}^{n=N} a_n \cos [k_n(x \cos \theta_n + y \sin \theta_n) - \omega_n t + \varphi_n]. \quad (1.18)$$

The direction θ_n and phase φ_n cover the range $-\pi, \pi$, and wave amplitudes and frequencies are from the range $0 \leq a_n \leq \infty$ and $0 \leq \omega_n \leq \infty$.

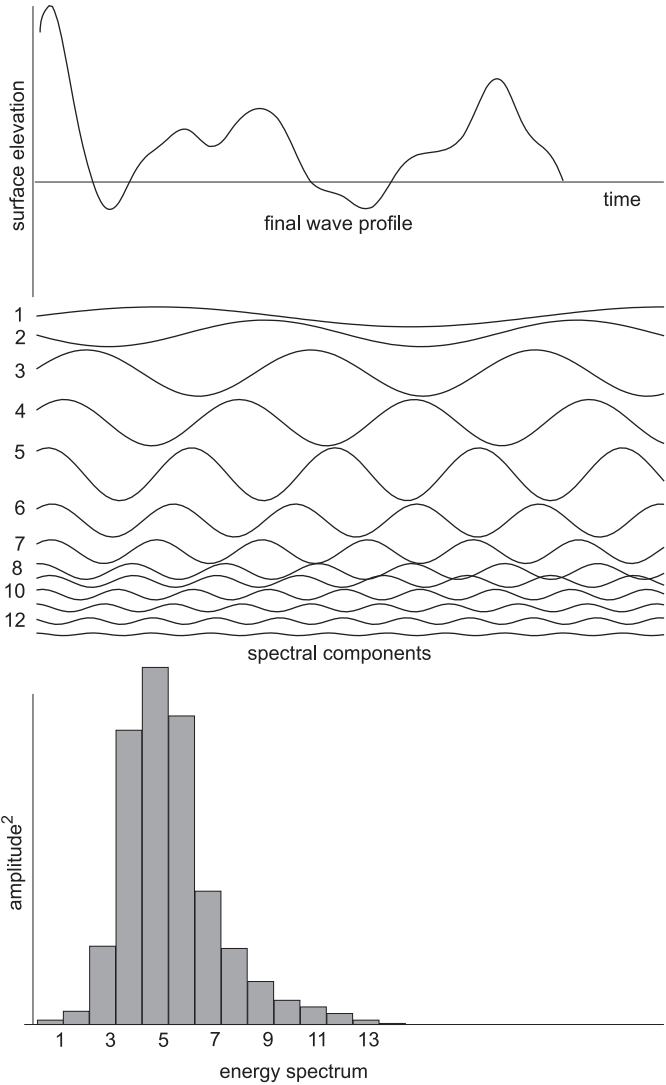


Fig. 1.2 Superposition of 13 spectral components and resulting spectrum.

The integral version of Eq. (1.18) would be

$$\zeta(x,y,t) = 2 \int_0^\infty \int_{-\pi}^\pi a(\omega,\theta) \cos [k(x \cos \theta + y \sin \theta) - \omega t + \varphi(\omega,\theta)] \, d\omega d\theta, \tag{1.19}$$

under the assumption that (Yaglom, 1962; Borgman, 1972)

$$a(\omega, \theta) = \frac{1}{2} \sum_{n=1}^N a_n \delta(\omega - \omega_n) \delta(\theta - \theta_n), \quad (1.20)$$

where $\delta(x)$ denotes Dirac' delta function⁵ (Lighthill, 1975). Using Euler's identity

$$\cos \alpha = \frac{1}{2} [\exp(i\alpha) + \exp(-i\alpha)], \quad (1.21)$$

we rewrite Eq. (1.19) in the form

$$\begin{aligned} \zeta(x, y, t) = \Re \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} a(\omega, \theta) \exp[i\varphi(\omega, \theta)] \\ \times \exp[ik(x \cos \theta + y \sin \theta) - \omega t] d\omega d\theta. \end{aligned} \quad (1.22)$$

Let us define the following cumulative function

$$A(\omega, \theta) = \int_{-\infty}^{\omega} \int_{-\pi}^{\theta} a(\omega', \theta') \exp[i\varphi(\omega', \theta')] d\omega' d\theta'. \quad (1.23)$$

Then, Eq. (1.22) becomes

$$\zeta(x, y, t) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] dA(\omega, \theta), \quad (1.24)$$

where $dA(\omega, \theta)$ is a differential element in the two-dimensional space (ω, θ) , $\mathbf{k}$ is a two-dimensional vector $(k \cos \theta, k \sin \theta)$ and $\mathbf{x}$ is the vector (x, y) .

For later convenience it will be useful to express Eq. (1.24) as a Fourier transform in three dimensions, (x, y, t) . Thus, we define

$$u = k \cos \theta, \quad v = k \sin \theta, \quad w = -\omega \quad (1.25)$$

and

$$\hat{A}(u, v, w) = \int_{-\infty}^u \int_{-\infty}^v \int_{-\infty}^w \hat{a}(u', v', w') dw' dv' du', \quad (1.26)$$

in which

$$\hat{a}(u, v, w) = A(\omega, \theta) \delta \left[|\omega| - (gk \tanh(kh))^{1/2} \right]. \quad (1.27)$$

⁵The Dirac's delta has two useful properties

$$\delta(x - x_0) = \begin{cases} \infty & \text{when } x = x_0 \\ 0 & \text{when } x \neq x_0 \end{cases}$$

and

$$\int_{-\infty}^{\infty} f(y) \delta(x - y) dy = f(x).$$

Using Eq. (1.25), the Eq. (1.24) may be rewritten as

$$\zeta(x, y, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp[i(ux + vy + wt)] d\hat{A}(u, v, w). \quad (1.28)$$

Due to the restriction Eq. (1.27) imposed on frequency ω and wavenumber k , a triple integral in Eq. (1.28) is in fact a double integral only.

The advantage of the integral representation (1.28) is that it makes application of the technique of Fourier integrals possible. This means that for given measurements of $\zeta(x, y, t)$ over the (x, y, t) space, the function $\hat{a}(u, v, w)$ can be computed in principle by an inverse Fourier transform. It can be seen from Eq. (1.27) that $\hat{a}(u, v, w)$ has nonzero values only when (u, v, w) satisfy the dispersion relation (1.17).

1.3.2 Nonlinear description of waves

1.3.2.1 Second-order Stokes waves

Linear description of ocean waves given by Eqs. (1.18) or (1.28) properly reflects only ocean waves with a small steepness. For higher steepness of regular waves, the second-order solution was first developed by (Stokes, 1847, 1980). However, for waves being superposition of components of various steepnesses and phases, an extension of the Stokes solution, which can also capture the effects of water depth, and directional spreading, is required. Such second-order solution for the infinite water depth was given by Longuet-Higgins (1963), and later by Sharma and Dean (1979) for arbitrary water depth. Their description was subsequently reproduced by Forristall (2000) as follows

$$\zeta(x, t) = \zeta^{(1)}(x, t) + \zeta^{(2)}(x, t), \quad (1.29)$$

in which

$$\zeta^{(1)}(x, t) = \sum_{n=1}^{n=N} a_n \cos(\mathbf{k}_n \cdot \mathbf{x} - \omega_n t + \varphi_n) \quad (1.30)$$

and

$$\zeta^{(2)}(x, t) = \frac{1}{4} \sum_{i=1}^N \sum_{j=1}^N a_i a_j \left[K^{(-)} \cos(\psi_i - \psi_j) + K^{(+)} \cos(\psi_i + \psi_j) \right], \quad (1.31)$$

where

$$K^{(-)} = \frac{D_{ij}^{(-)} - (\mathbf{k}_i \cdot \mathbf{k}_j + R_i R_j)}{\sqrt{R_i R_j}} + (R_i + R_j), \quad (1.32)$$

$$K^{(+)} = \frac{D_{ij}^{(+)} - (\mathbf{k}_i \cdot \mathbf{k}_j - R_i R_j)}{\sqrt{R_i R_j}} + (R_i + R_j), \quad (1.33)$$

$$D_{ij}^{(-)} = \frac{(\sqrt{R_i} - \sqrt{R_j}) [\sqrt{R_j}(\mathbf{k}_i^2 - R_i^2) - \sqrt{R_i}(\mathbf{k}_j^2 - R_j^2)]}{(\sqrt{R_i} - \sqrt{R_j})^2 - k_{ij}^{(-)} \tanh(k_{ij}^{(-)} h)} + \frac{2(\sqrt{R_i} - \sqrt{R_j})^2 (\mathbf{k}_i \cdot \mathbf{k}_j + R_i R_j)}{(\sqrt{R_i} - \sqrt{R_j})^2 - k_{ij}^{(-)} \tanh(k_{ij}^{(-)} h)}, \quad (1.34)$$

$$D_{ij}^{(+)} = \frac{(\sqrt{R_i} + \sqrt{R_j}) [\sqrt{R_i}(\mathbf{k}_j^2 - R_j^2) + \sqrt{R_j}(\mathbf{k}_i^2 - R_i^2)]}{(\sqrt{R_i} + \sqrt{R_j})^2 - k_{ij}^{(+)} \tanh(k_{ij}^{(+)} h)} + \frac{2(\sqrt{R_i} + \sqrt{R_j})^2 (\mathbf{k}_i \cdot \mathbf{k}_j - R_i R_j)}{(\sqrt{R_i} + \sqrt{R_j})^2 - k_{ij}^{(+)} \tanh(k_{ij}^{(+)} h)}, \quad (1.35)$$

$$k_{ij}^{(-)} = |\mathbf{k}_i - \mathbf{k}_j|, \quad k_{ij}^{(+)} = |\mathbf{k}_i + \mathbf{k}_j|, \quad (1.36)$$

$$R_i = |\mathbf{k}_i| \tanh(|\mathbf{k}_i| h) = \omega_i^2 / g, \quad \psi_i = \mathbf{k}_i \cdot \mathbf{x} - \omega_i t + \varphi_i. \quad (1.37)$$

For infinite water depth, Eq. (1.31) reduces to the result of Longuet-Higgins (1963), except that the latter equation is missing a factor of $1/2$.

The double sum in Eq. (1.31) contains all the second-order bound waves that are generated by interactions between the free waves.⁶ The positive interaction terms $(\psi_i + \psi_j)$ produce the sharpening of the crests and flattening of the troughs. The negative interaction terms $(\psi_i - \psi_j)$ yield the set-down of the water level under wave groups. It is clearly seen in Fig. 1.3 that the bound waves do not satisfy the linear dispersion relation, as

$$(\omega_i \pm \omega_j) \sqrt{\frac{h}{g}} \neq (k_i \pm k_j) \tanh[(k_i \pm k_j) h]. \quad (1.38)$$

At present, the second-order wave description is widely used in the offshore industry (Birknes and Bitner-Gregersen, 2003; Bitner-Gregersen and Hagen, 2004).

⁶It should be noted that the wave energy spectrum does not change in time as the wave field evolves. Therefore in the second-order solution, the nonlinear dynamics of free waves and the modulational instability are not included.

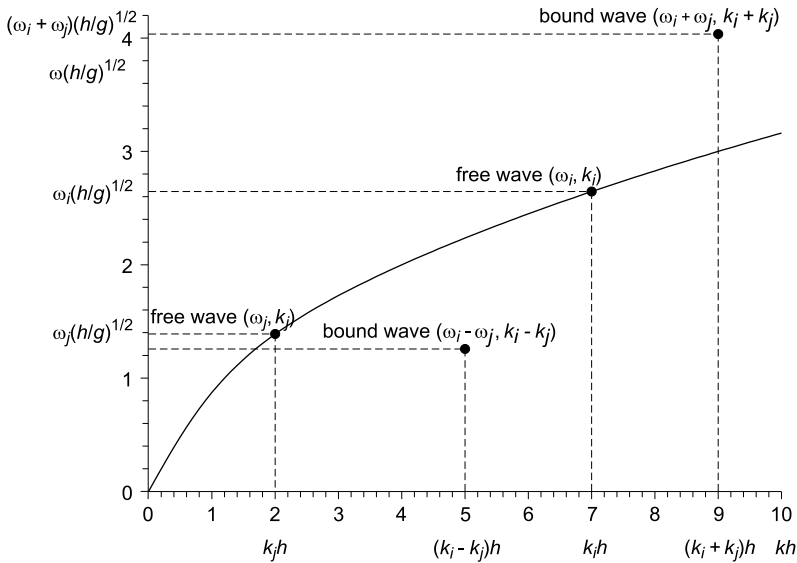


Fig. 1.3 Dispersion relation for free and bound waves.

1.3.2.2 Higher-order Stokes waves

For single regular wave profile, the Stokes type solution to a higher order can be found. If we assume the regular, plane wave propagating over constant water depth in x -direction, we obtain the following fifth-order wave surface profile (Fenton, 1985)

$$\begin{aligned} \zeta(x, t) = & \frac{1}{k} \left\{ \varepsilon \cos \psi + \varepsilon^2 B_{22} \cos 2\psi + \varepsilon^3 B_{31} (\cos \psi - \cos 3\psi) \right. \\ & + \varepsilon^4 (B_{42} \cos 2\psi + B_{44} \cos 4\psi) + \varepsilon^5 [-(B_{53} + B_{55}) \cos \psi \\ & \left. + B_{53} \cos 3\psi + B_{55} \cos 5\psi] \right\} + O(\varepsilon^6), \end{aligned} \quad (1.39)$$

where $\varepsilon = k(H/2)$ and $\psi = kx - \omega t$. The set of coefficients B_{ij} is given in Fenton (1985).

The fifth-order Stokes waves are often used for assessment of the maximum surface displacement and maximum water velocity and acceleration beneath the high wave crest. The velocity and acceleration are required for drag force and inertia force calculations. Therefore, for completeness we provide the formulae for the horizontal and vertical velocity components,

as well as for the Eulerian and Lagrangian accelerations (Fenton, 1985):

– horizontal velocity component

$$u(x, z, t) = C_0 \left(\frac{g}{k^3} \right)^{1/2} \sum_{i=1}^5 \sum_{j=1}^i A_{ij} \varepsilon^i jk \cosh[jk(z+h)] \cos(j\psi) \quad (1.40)$$

– vertical velocity component

$$w(x, z, t) = C_0 \left(\frac{g}{k^3} \right)^{1/2} \sum_{i=1}^5 \sum_{j=1}^i A_{ij} \varepsilon^i jk \sinh[jk(z+h)] \sin(j\psi), \quad (1.41)$$

where

$$\psi = kx - \omega t. \quad (1.42)$$

The wavenumber k is given by the formula

$$-\frac{2\pi}{T\sqrt{gk}} + C_0(kh) + \varepsilon^2 C_2(kh) + \varepsilon^4 C_4(kh) = 0. \quad (1.43)$$

The coefficients A_{ij} and C_i can be found in Fenton (1985).

Using the above formulae we obtain the Eulerian $a_z^{(E)}$ and Lagrangian $a_z^{(L)}$ accelerations as follows

$$a_z^{(E)} = \frac{\partial^2 \zeta(x, t)}{\partial t^2} = \frac{\partial w(x, \zeta, t)}{\partial t} \quad (1.44)$$

and

$$a_z^{(L)}(x, \zeta, t) = \frac{Dw}{Dt} = \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + w \frac{\partial w}{\partial z} = \frac{\partial w}{\partial t} + u \frac{\partial u}{\partial z} - w \frac{\partial u}{\partial x} \text{ at } z = \zeta. \quad (1.45)$$

For illustration, in Fig. 1.4, the comparison of the surface displacements for linear, second-order and fifth-order approximation is given with the following wave parameters used in calculations: water depth $h = 20$ m, wave length $L = 250$ m, wave height $H = 6$ m, $H/h = 0.3$ and $L/h = 12.5$.

1.3.2.3 Nonlinear Schrödinger (NLS) equations and modulational instability

In the Stokes approximations, even as high as the fifth-order, the evolution of waves in time or in space is not included and the energy spectrum remains constant. However, experiments in both natural and laboratory conditions demonstrate clearly that wave train in general is unstable and undergoes modulations due to nonlinear character of wave motion. When

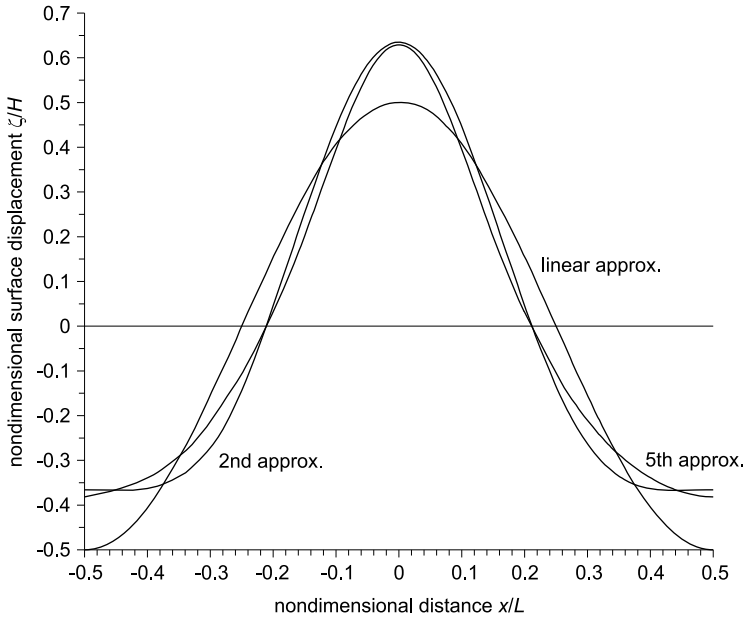


Fig. 1.4 Comparison of linear and higher order approximations of surface displacement.

the ocean surface waves are regarded as narrow spectrum waves, the non-linear evolution of the wave train in time is studied by using the cubic Nonlinear Schrödinger (NLS) equation (Peregrine, 1983a; Kharif and Pelinovsky, 2003; Mei *et al.*, 2005; Onorato *et al.*, 2006). The NLS equation describing the complex envelope of sea elevations for arbitrary water depth (in a frame of reference moving with the group velocity) is

$$\frac{\partial A}{\partial t} + i\sigma \frac{\omega_p}{8k_p^2} \frac{\partial^2 A}{\partial x^2} + i\beta \frac{\omega_p k_p^2}{2} |A|^2 A = 0, \quad (1.46)$$

where k_p is the carrier wavenumber, ω_p is the respective angular frequency and σ and β are functions of the nondimensional wavenumber $k_p h$

$$\sigma = 2 - \nu_1^2 + 8(k_p h)^2 \frac{\cosh(2k_p h)}{\sinh^2(2k_p h)}, \quad (1.47)$$

$$\beta = \frac{8 + \cosh(4k_p h) - 2 \tanh^2(k_p h)}{8 \sinh^4(k_p h)} - \frac{[2 \cosh^2(k_p h) + 0.5 \nu_1]^2}{\sinh^2(2k_p h)(k_p h) / \tanh(k_p h) - \nu_1^2 / 4} \quad (1.48)$$

and

$$\nu_1 = 1 + 2 \frac{k_p h}{\sinh(2k_p h)}. \quad (1.49)$$

When $k_p h \rightarrow \infty$, coefficients σ and β tend to 1.

The complex amplitude A is a slowly varying function of x and t with a third-order accuracy in the wave slope ka , and the surface elevation $\zeta(x, t)$ is given by

$$\zeta(x, t) = \frac{1}{2} (A(x, t) \exp[i(k_p x - \omega_p t)] + c.c.), \quad (1.50)$$

where *c.c.* denotes the complex conjugate.

The modulational instability leads to decomposition of an initially homogeneous wave train into a system of envelope quasi-solitons, and the spatial distribution of wave energy displays significant intermittency (see Fig. 1.5).

The wave steepness and wave dispersion play a critical role for an evolution and stability of the water waves (Benjamin and Feir, 1967). The popular measure of the relative importance of nonlinearity and wave dispersion is so called Benjamin–Feir index (*BFI*), introduced initially by Janssen (2003). To develop the Benjamin–Feir index we define following Onorato *et al.* (2001, 2006) the nondimensional quantities: $\tilde{A} = A/a_0$, $\tilde{x} = x/\Delta K$, $\tilde{t} = t(\Delta K/k_p)^2 \sigma \omega_p / 8$, where ΔK is a typical spectral band-width and a_0 is a typical wave amplitude.

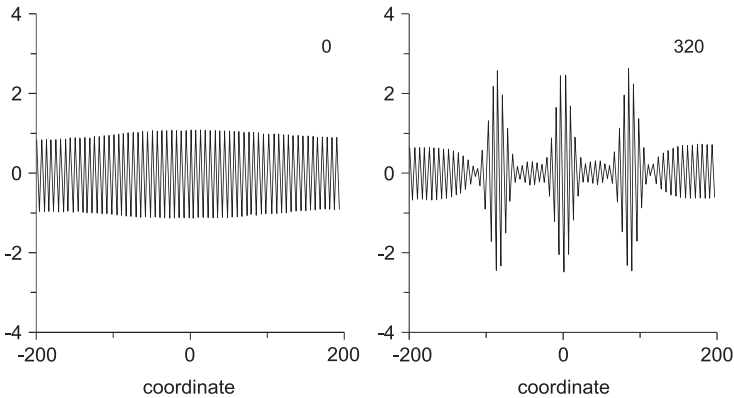


Fig. 1.5 Formation of a highly energetic wave group in a slowly modulated wave train. The time (0 on the left and 320 on the right) is normalised by the fundamental wave period and the coordinate is in radius of the fundamental mode (adapted from Kharif and Pelinovsky, 2003).

After substituting $\tilde{A}$, $\tilde{x}$ and $\tilde{t}$ into Eq. (1.46) we obtain

$$\frac{\partial \tilde{A}}{\partial \tilde{t}} + i \frac{\partial^2 \tilde{A}}{\partial \tilde{x}^2} + i \left(\frac{2k_p a_0}{\Delta K / k_p} \right)^2 \frac{\beta}{\sigma} |\tilde{A}|^2 \tilde{A} = 0. \quad (1.51)$$

The Benjamin–Feir index BFI is defined as the square root of the coefficient that multiplies the nonlinear term in above equation, i.e.

$$BFI = \frac{2k_p a_0}{\Delta K / k_p} \sqrt{\frac{|\beta|}{\sigma}}. \quad (1.52)$$

The term $\sqrt{|\beta|/\sigma}$ represents the influence of the water depth. When water depth increases, this term goes to one and for shallow water depth it becomes zero. For values of $k_p a_0 < 1.36$, the coefficient in front of the nonlinear term in the NLS equation changes sign and the equation becomes stable with respect to side perturbations.

As the BFI increases the nonlinearity increases, and the maximum instability occurs when $\Delta K / k_p = 2k_p a_0$. Therefore for wave field sufficiently steep, narrow banded, and unidirectional, the random waves become unstable when $BFI = \mathcal{O}(\infty)$. If the time series (or frequency spectra) are available, the following definition of BFI is used

$$BFI = \frac{(k_p a_0)}{\Delta \omega / \omega_p} \sqrt{\frac{|\beta|}{\sigma}}, \quad (1.53)$$

which can be rewritten in terms of the significant wave height H_s and peak wavenumber k_p (Bitner-Gregersen and Toffoli, 2014)

$$BFI = \frac{k_p H_s}{2\Delta \omega / \omega_p}. \quad (1.54)$$

The computation methods of the BFI for a given time series or wave spectrum were considered by Serio *et al.* (2005). For more information on the Benjamin–Feir instability, the reader should consult Benjamin (1967), Krasitskii (1994) and Janssen (2003).

Dysthe (1979) extended the cubic NLS equation to fourth order in waves steepness and bandwidth which can be solved more easily by a pseudospectral numerical method (Lo and Mei, 1985). Beside the Dysthe extension, there are numerous other variants of NLS-type equations for special conditions (Trulsen and Dysthe, 1997; Onorato *et al.*, 2000, 2001, 2006; Kharif and Pelinovsky, 2003; Kurkin and Pelinovsky, 2003; Toffoli *et al.*, 2007, 2008a, 2010b, 2011, 2016).

In the laboratory practice, wave data are usually collected from sensors located at fixed positions in the wave basin. Therefore for comparison

of such data with numerical model predictions, the governing equations have to be transferred to the spatial form (Lo and Mei, 1985). The exact expressions for the temporal and spatial modified NLS equations (MNLS) for complex amplitude $A(x, y, t)$ of the directional wave field take the form (Socquet-Juglard *et al.*, 2005; Toffoli *et al.*, 2010b; Zhang *et al.*, 2016)

$$\begin{aligned} & \frac{\partial A}{\partial t} + \frac{1}{2} \frac{\partial A}{\partial x} + \frac{i}{8} \frac{\partial^2 A}{\partial x^2} - \frac{i}{4} \frac{\partial^2 A}{\partial y^2} + \frac{i}{2} |A|^2 A \\ & - \frac{1}{16} \frac{\partial^3 A}{\partial x^3} + \frac{3}{8} \frac{\partial^3 A}{\partial x \partial y^2} + \frac{3}{2} |A|^2 \frac{\partial A}{\partial x} + \frac{1}{4} A^2 \frac{\partial A^*}{\partial x} \\ & + iA \frac{\partial \bar{\phi}}{\partial x} = 0, \quad z = 0 \end{aligned} \quad (1.55)$$

and

$$\begin{aligned} & \frac{\partial A}{\partial x} + 2 \frac{\partial A}{\partial t} + i \frac{\partial^2 A}{\partial t^2} - \frac{i}{2} \frac{\partial^2 A}{\partial y^2} + i |A|^2 A - \frac{\partial^3 A}{\partial t \partial y^2} \\ & - 8 |A|^2 \frac{\partial A}{\partial t} - 2 A^2 \frac{\partial A^*}{\partial t} - 4 i A \frac{\partial \bar{\phi}}{\partial t} = 0, \quad z = 0, \end{aligned} \quad (1.56)$$

where A^* is the complex conjugate of wave amplitude A .

The main shortcomings of NLS equation are the requirements of narrow wave spectrum and slow modulation of waves, which limits its practical applicability. A more general alternatives to NLS equations are so-called Zakharov equations (Zakharov, 1968), which assume weak nonlinearity but with no assumptions on the spectral bandwidth. In practical computations, the Zakharov equation is discretised in terms of a fixed number of free wave modes which are subject to its own evolution equations. An important computational method for solving the Zakharov equation was formulated by West *et al.* (1987) and Dommermuth and Yue (1987) as the Higher Order Spectral Method (HOSM). This relatively efficient numerical method is a pseudo-spectral type method where a series expansion in the wave slope of the vertical velocity about the free surface is used. The method does not contain any additional assumptions, except for the surface being single-valued (i.e. no wave breaking). High efficiency and accuracy makes HOSM a good tool for studying the nonlinear wave dynamics. Mei *et al.* (2005) described in detail mathematical formulation and numerical implementation of the HOSM for simulation of nonlinear wave-wave, wave-current and wave-bottom interactions. The method has been subsequently extended to include the presence of atmospheric forcing, finite depth variations, variable current and moving submerged bodies. In recent years, the

HOSM has been applied by DNV GL (Norway) for studies on the rogue waves (Bitner-Gregersen and Gramstad, 2015). For more detail review of HOSM, the reader should consult Tanaka (2001).

Further extension of fully nonlinear wave modelling was reported in a series of papers by Bateman *et al.* (2001, 2003, 2012). The developed model for realistic ocean spectra incorporates spread of wave energy across the frequency and directional domains, well as the evolution of the largest waves. This procedure provides full-field (both spatial and temporal) description of the evolving water surface $\zeta(x, y, t)$, together with the associated water particle kinematics $\mathbf{u}(x, y, z, t)$. Given a valid set of initial conditions, the applied calculation procedure is essentially exact and has been used to provide sample calculations involving the JONSWAP spectra in relatively deep water. However, model is not restricted to this flow regime and can be successfully applied for any water depths. Although the modified nonlinear Schrödinger equations can be employed to describe the modulations of a regular and irregular wave trains over some wave cycles, they are not appropriate to the description of extreme waves arising in broad-band sea states.

The engineering practise frequently requires the information concerning very nonlinear ocean waves, including the limiting breaking waves. For unidirectional waves propagating in a two-dimensional fluid, the time-marching procedure allowing the evolution of the wave-field has been developed using the inverse method in which potential function φ and stream function ψ are adopted as the dependent variables rather than coordinates (x, z) of the common physical plane method (Longuet-Higgins and Cokelet, 1976; Cokelet, 1977). The solution is valid to the limiting breakers. However, this type of solution is seldom applied in engineering design because it can not easily be expanded to include the effects of directional spreading and it requires significant computing resources.

The wave kinematics and loads due to extreme waves are very important for offshore engineering operations, offshore structures and ships performance. An accurate assessment of the maximum water particle velocities beneath the high wave crest is required for drag force calculations, while the maximum accelerations beneath the steepest section of the wave profile are critical for inertial force calculations. The wave models used to determine the water particle kinematics associated with a measured time history of the surface elevation $\zeta(t)$ are traditionally based either on a nonlinear regular wave theory (for example a higher-order Stokes' theory) or on an unsteady linear wave theory (a Fourier transform approach).

This is not the case for rough waves and near breaking waves when the kinematics is not well predicted by any existing theory. A fully non-linear unsteady wave model has been developed by Clamond and Grue (2001a, b) which offers a substantial improvement in the modelling of steep wave kinematics. The model is based on potential theory and assumes the integration of the kinematic and dynamic boundary conditions at the free surface of very steep (rough) waves. The theoretical results compare favourably with various available sets of experimental data. Measurements and computations show some deep water waves with fluid velocity up to 75% of the phase speed (Grue *et al.*, 2003). The PIV and LDA data demonstrate excellent agreement with the fully non-linear theory. The experimental data correspond to the JONSWAP spectrum with $\omega_p^2 H_s / 2g = 0.15$ and the range of the wave slope is $0.21 < \varepsilon_s < 0.34$. A surprising result is that the exponential profile e^{kz} compares well with all measurements in deep water.

Accelerations at the sea surface are required to estimate the inertia forces on offshore installations. Very few measurements exist of the real acceleration below steep waves (see for example Bonmarin and Kjeldsen, 2000; Grue and Jensen, 2006). In particular, Grue and Jensen reported a maximum negative vertical acceleration value at wave crest of about $1.1g$, while the upward vertical acceleration grows to about $1.5g$ in the front face of the wave at the base below the overturning jet of the plunging breakers. The fully non-linear theoretical model for accelerations in unsteady waves shows very good agreement with experiments. It was shown that the convective term is of the same order of magnitude and of opposite sign to the local acceleration, and it cannot be neglected when estimating the acceleration and forces (Jensen *et al.* 2007).

1.3.2.4 Boussinesq and Korteweg-de Vries equations for waves in shallow water

In the coastal zone, the typical wave length, L , is much larger than water depth h . Moreover, the wave height, H , is an appreciable fraction of h , so the Ursell number becomes too large to allow the Stokes approach to be used. This is also true for tsunami waves approaching the coast or for the internal waves (Massel, 2015).

Let us define two small parameters δ and ν :

- relative wave length $\delta = kh$
- relative wave amplitude $\nu = A/h$.

Therefore the Ursell number U becomes

$$U = \frac{H}{h} \left(\frac{L}{h} \right)^2 \quad (1.57)$$

or

$$U = 2(2\pi)^2 \frac{\frac{A}{\bar{h}}}{(kh)^2} \approx 75 \frac{\nu}{\delta^2}. \quad (1.58)$$

The parameters δ and ν allow to distinguish between three different important models of the long waves:

- a) nonlinear shallow water equation
 $\nu \gg \delta^2, \quad \nu = \mathcal{O}(1), \quad U \gg 75$
- b) Boussinesq and Korteweg–de Vries equation
 $\mathcal{O}(\nu) \approx \mathcal{O}(\delta^2) < 1, \quad U \sim \mathcal{O}(75)$
- c) linearised long wave equation
 $\nu \ll \delta^2, \quad U \ll 75.$

In the following we will further consider the case of the Boussinesq and Korteweg–de Vries equations.⁷ The essence of the method related to these equations lies in eliminating the z -dependence and adopting the characteristic horizontal velocity either as the velocity at $z = 0$ or at the bottom $z = -h$ or the velocity averaged over the depth. When we use the horizontal velocity $\bar{u}$ averaged over the water depth, we obtain the Boussinesq equations as follows (Whitham, 1974)

$$\left. \begin{aligned} \frac{\partial \zeta}{\partial t} + \frac{\partial}{\partial x} [(h + \zeta)\bar{u}] &= 0 \\ \frac{\partial \bar{u}}{\partial t} + \bar{u} \frac{\partial \bar{u}}{\partial x} + g \frac{\partial \zeta}{\partial x} &= \frac{1}{3} h^3 \frac{\partial^3 \bar{u}}{\partial x^2 \partial t} \end{aligned} \right\}. \quad (1.59)$$

For waves propagating in the given direction, Eq. (1.59) is reduced to the Korteweg–de Vries (KdV) equation (Korteweg and de Vries, 1895). The KdV equation in terms of the surface elevation takes the form

$$\frac{\partial \zeta}{\partial t} + \sqrt{gh} \left(1 + \frac{3}{2} \frac{\zeta}{h} \right) \frac{\partial \zeta}{\partial x} + \frac{h^2}{6} \sqrt{gh} \frac{\partial^3 \zeta}{\partial x^3} = 0, \quad (1.60)$$

⁷Joseph Valentin Boussinesq (1842–1929) was a French mathematician and physicist who made significant contributions to the theory of hydrodynamics, vibration and heat. He developed the mathematical theory to support Russell's observations of the solitary waves, and introduced the KdV equation in 1877 which was rediscovered by Diederik Korteweg and Gustaw de Vries in 1895.

while in terms of the velocity $\bar{u}$ we have

$$\frac{\partial \bar{u}}{\partial t} + \left(\sqrt{gh} + \frac{3}{2} \bar{u} \right) \frac{\partial \bar{u}}{\partial x} + \frac{h^2}{6} \sqrt{gh} \frac{\partial^3 \bar{u}}{\partial x^3} = 0. \quad (1.61)$$

The KdV equation describes the evolution of weakly dispersive and weakly nonlinear waves. It has the periodic and stable solutions, namely the solitary waves and cnoidal waves. In particular, the surface elevation of the solitary wave for small value of ν parameter is

$$\zeta(x, t) = H \cosh^{-2} \left(\sqrt{\frac{3H}{4h^3}} (x - C_p t) \right), \quad (1.62)$$

in which

$$C_p = \sqrt{gh} \left(1 + \frac{H}{h} \right)^{1/2} \approx \sqrt{gh} \left[1 + \frac{1}{2} \left(\frac{H}{h} \right) \right]. \quad (1.63)$$

The cnoidal wave form of the surface displacement resulting from the KdV equation provides an opportunity to study dynamics of the simple linear sinusoidal waves as well as solitary waves within the same framework of the Jacobian elliptic functions (Abramowitz and Stegun, 1975). Thus the cnoidal wave shape becomes (Massel, 1989)

$$\zeta(x, t) = H(x) \left\{ \frac{1}{m} \left(1 - \frac{\mathbf{E}}{\mathbf{K}} - m \right) + cn^2 \left[\frac{2\mathbf{K}}{L} (x - C_p t) \right] \right\}, \quad (1.64)$$

in which $H(x)$ is the wave height, L is the wave length and $cn(x)$ is the Jacobian elliptic function, $\mathbf{K}$ and $\mathbf{E}$ are the complete elliptic integrals of the first and second kind, respectively. They are functions of the elliptic parameter m from the range $0 \leq m \leq 1$ (Abramowitz and Stegun, 1975). The phase speed C_p is given by

$$C_p^2 = gh \left[1 + \frac{H}{h} \frac{1}{m} \left(2 - 3 \frac{\mathbf{E}}{\mathbf{K}} - m \right) \right]. \quad (1.65)$$

To define an elliptic parameter m , the following relationship is used

$$\left(\frac{H}{h} \right) \left(\frac{L}{h} \right)^2 = \frac{16}{3} m \mathbf{K}^2. \quad (1.66)$$

It should be noted that if $m = 0$, the Eq. (1.64) becomes

$$\zeta(x, t) = \frac{H}{2} \cos \frac{2\pi}{L} (x - C_p t), \quad (1.67)$$

which is a linear sinusoidal wave. For elliptic parameter $m = 1$, a limit of the cnoidal wave is the solitary wave of infinite wave length — see Eq. (1.62). For more systematic review of the solitary and cnoidal waves the reader

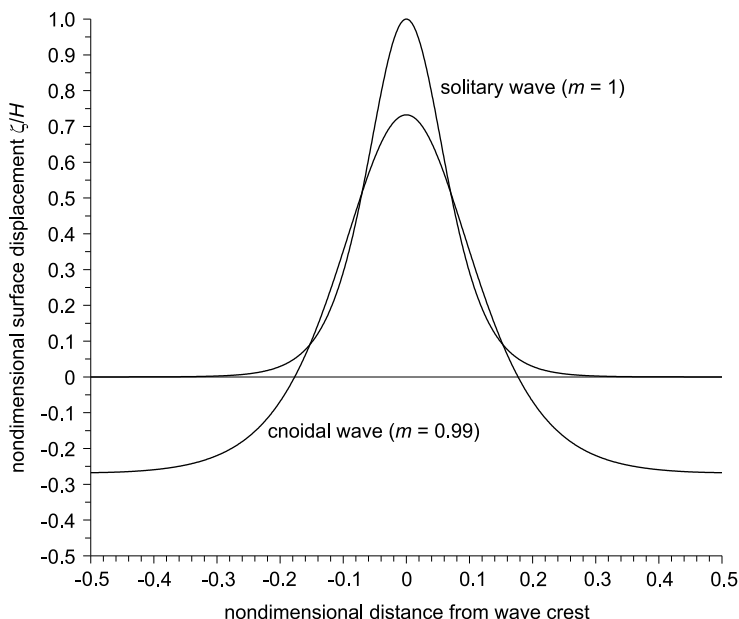


Fig. 1.6 Comparison of solitary and cnoidal wave displacements.

should refer to the paper by Fenton (1979). For illustration, the comparison of the solitary and cnoidal wave profiles is shown in Fig. 1.6, in which the cnoidal profile corresponds to the elliptic parameter $m = 0.99$.

There are several proposals for the range of application of various wave theories. The applicability of the second order Stokes' perturbation method and cnoidal wave theory for determination of the limiting wave height in shallow water is considered in some detail in Section 7.2.8. For the practical purposes, the ranges of validity of wave theories has been given by Druet (1978) and Chakrabarti (1987) in terms of wave shallowness (h/T^2) and wave steepness (H/T^2).

1.4 Methods of Description of Ocean Random Waves

1.4.1 Preliminaries

In looking at ocean surface waves one notices both their randomness and their quasi-regularity. The wave profile changes constantly with time and in a random fashion. Consequently, the properties of waves are not readily

time series $\zeta(t)$	time domain stationary process - autocorrelation function $K(\tau)$ non-stationary process - Hilbert Transform $H[\zeta(t)]$
	frequency domain frequency spectra
	time-frequency domain wavelets
	probability domain random values statistics

Fig. 1.7 Methods of processing of random time series $\zeta(t)$.

defined on a wave-by-wave basis. As the fundamental property of surface waves, induced by wind, is their irregularity, the prediction of wave parameters can be achieved only through stochastic analysis of the sea data, which span three basic domains, namely frequency domain, probability domain, and time domain, as was demonstrated in Fig. 1.7 for random time series $\zeta(t)$.

Frequency analysis deals mainly with an evaluation of the distribution of wave energy (potential or kinetic) among various frequencies and directions. There are two main methods for the development of the frequency spectrum. The traditional method is based on the Fourier transform of the auto-correlation function which is a measure of the connection between the two values $\zeta(t)$ and $\zeta(t + \tau)$ of the random variable ζ . The theoretical background of this transformation is provided by the Wiener–Khinchine Theorem. The transformation of the auto-correlation function gives the spectral density function of the particular variable, and the transformation of cross-correlation function results in cross-spectral density function, which is a complex function. The real part of the cross-spectral function is called the co-spectrum and the imaginary part is called the quadrature spectrum. A more general spectral representation of surface waves is obtained when the frequency, as well the directional energy distribution, is taken into account. The resulting spectrum is called the frequency-directional spectrum.

The second spectral method is based on the straightforward transferring of the time series into its Fourier components. This technique, known as Fast Fourier Transform (FFT), was first introduced in a new form by Cooley and Tukey (1965). It reduces the number of calculations from a number proportional to n^2 (n is a number of samples) to a number approximately proportional to $n \log n$ and has revolutionised the spectral analysis of time series. One major weakness of the Fourier-based spectral methods is the assumption of linear superposition of wave components, which, in fact, are phase-coupled via the nonlinear dynamics present in ocean waves. It should be noted that the Fourier transform provides an information on the frequency content, however, the information on the frequency localisation in time is essentially lost in the process. In addition, Fourier spectral analysis should, strictly speaking, be used for periodic and stationary processes only. When the time localisation of the spectral components is needed, the transform of time series which provides the time-frequency representation of the signal is the Wavelet transform (see Section 1.4.4.1).

In the probability domain, the particular wave parameters, e.g. surface displacements at a given time, wave amplitudes, wave heights, wave periods, etc. are considered as elementary random events. The probability approach is easy to understand when we are dealing with digitised data. The digitised data of a particular parameter form a set of random realisations of a random variable, when the time sequence of the parameter is exterminated. The final results in this approach are expressed in terms of probability density functions, distribution functions and moments (see Chapter 4).

The simplest statistics are obtain when one assumes that the observed wave field is a result of a linear superposition of a large number of dynamically independent waves. This is the basis of the Gaussian model, in which two first moments are sufficient for the complete statistical description of the wave field. However, in the real ocean, due to nonlinear interactions between spectral harmonics and energy dissipation processes, a substantial departure from the Gaussian model is observed. Ocean waves should be treated in many cases as example of the non-Gaussian stochastic processes.

For processing of the nonstationary time series within time domain, the Hilbert transform becomes very useful tool. The Hilbert transform of a real-valued time domain signal $\zeta(t)$ is another real-valued time domain signal $\xi(t)$ such that

$$\eta(t) = \zeta(t) + i\xi(t) = A(t) \exp(i\theta(t)), \quad (1.68)$$

where $\xi(t) = H[\zeta(t)]$ is the Hilbert transform of function $\zeta(t)$, and function $\eta(t)$ forms an analytical signal, also in time domain. This is a substantial

Table 1.2 A comparative summary of Fourier, Wavelet and Hilbert transforms methodology (adapted from Huang and Shen, 2005).

	Fourier	Wavelet	Hilbert
Basis	<i>a priori</i>	<i>a priori</i>	adaptive
Frequency	convolution global uncertainty	convolution regional uncertainty	differentiation local, certainty
Presentation	energy- frequency	energy-time- frequency	energy-time- frequency
Nonlinear	no	no	yes
Nonstationary	no	yes	yes
Feature Extraction	no	discrete no; continuous yes	yes
Theoretical base	theory complete	theory complete	empirical

difference when comparing with the Fourier transform which transforms signal $\zeta(t)$ into a complex-valued frequency domain signal. Representation (1.68) provides the best local fit of an amplitude and phase for function $\zeta(t)$ varying in time, as well as give some information on the instantaneous frequency nature. A comparative summary of Fourier transform, Wavelet transform and Hilbert transform methods are given in Table 1.2.

1.4.2 Basic definitions and analysis of random time series

Let us begin with an ensemble of k wave records $\{\zeta_k(t)\}$ taken under identical macroscopic conditions, i.e. position on the ocean surface, water depth, mean wind velocity, air-sea temperature, etc. Even under identical conditions, we cannot expect that these wave records will be identical or even closely similar in detail. The family $\{\zeta_k(t)\}$ represents k realisations of the stochastic process $\zeta(t)$. For a given k , $\zeta(t)$ is a function of time t , while when $t = t_1$, $\zeta_k(t_1)$ is a random variable.

Stochastic processes may belong to one of three categories: a) stationary and ergodic, b) stationary, and c) non-stationary. Random process (or random function) is stationary in the *wide sense* if

$$E[\zeta(t)] = \bar{\zeta} = \text{const},$$

(1.69)

and

$$K(t_1, t_2) = K(t_1 - t_2) = E[\zeta(t_1) \zeta(t_2)] = K(\tau), \quad \tau = t_1 - t_2,$$

(1.70)

in which $E[\]$ denotes the mean of ζ and $K(\)$ is an auto-correlation function. Strictly speaking, a random process is stationary if all statistical moments are invariant. These two definitions of stationarity coincide when ζ is a Gaussian, in which case all the statistics of ζ are completely determined by the first and second moments. This strict definition is often relaxed and stationarity in wide sense is assumed.

In general, using the ensemble of wave records $\{\zeta_k(t)\}$, we can develop any function of ζ , say F , to give $F\{\zeta_k(t)\}$. To be more specific, we select time $t = t_1$ in the family $\{\zeta_k(t)\}$. When F is the ζ value itself, then averaging $F\{\zeta_k(t_1)\}$ over k results in the ensemble mean of the process at $t = t_1$, i.e.

$$E[F\{\zeta_k(t_1)\}]_k = E[\zeta_k(t_1)]_k = \lim_{N \rightarrow \infty} \frac{\sum_{k=1}^N \zeta_k(t_1)}{N}. \quad (1.71)$$

The condition $N \rightarrow \infty$ is only conceptual since N is usually finite.

When $F\{\zeta_k(t_1)\} \equiv [\zeta_k(t_1)]^2$, then averaging $F\{\zeta_k(t_1)\}$ over k leads to the variance at $t = t_1$

$$E[F\{\zeta_k(t_1)\}]_k = E\left\{[\zeta_k(t_1)]^2\right\}_k = \lim_{N \rightarrow \infty} \frac{\sum_{k=1}^N [\zeta_k(t_1)]^2}{N}. \quad (1.72)$$

Let us now define F as

$$F\{\zeta_k(t_1)\} = \begin{cases} 1 & \text{if } a < \zeta_k(t_1) \leq b \\ 0 & \text{otherwise.} \end{cases} \quad (1.73)$$

Averaging over the ensemble $E[F\{\zeta_k(t_1)\}]_k$ can be interpreted as the ensemble probability that $\zeta_k(t_1)$ falls on the interval from a to b at $t = t_1$.

Repetition of the above averaging for the different times helps us to obtain the different numerical values for the statistics. However, the repeated observations technique which provides us with an ensemble of k wave records can be attained in laboratory wave tanks, but is inapplicable to observations of wave phenomena in field experiments. To overcome these difficulties, the ergodic theorem is usually invoked. This allows the ensemble averages to be replaced with time averages. The ergodic theorem states that (Ochi, 1998): *if $\zeta(t)$ is an ergodic stationary random function, then the statistics obtained by ensemble averages at a given time $t = t_*$ are identical to the corresponding statistics computed by the time averaging for any given realisation $k = k_*$.* Hence, the ergodic stationary process should satisfy the

following equality for record of time T

$$\begin{aligned} E[F\{\zeta_k(t=t_*)\}]_k &= \lim_{N \rightarrow \infty} \frac{\sum_{k=1}^N F\{\zeta_k(t=t_*)\}}{N} = E[F\{\zeta_{k=k_*}(t)\}]_t \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T F\{\zeta_{k=k_*}(t)\} dt. \end{aligned} \quad (1.74)$$

If above condition is limited to the mean and covariance, then the stochastic process is called a *weakly ergodic random process*. Therefore we have

$$\frac{1}{N} \sum_{k=1}^N x_k(t) = \lim_{T \rightarrow \infty} \int_0^T x(t) dt = \bar{x} \quad (1.75)$$

and

$$\frac{1}{N} \sum_{k=1}^N [x_k(t) - \bar{x}][x_k(t+\tau) - \bar{x}] = \lim_{T \rightarrow \infty} \int_0^T [x(t) - \bar{x}][x(t+\tau) - \bar{x}] dt. \quad (1.76)$$

We can say that, while stationary processes are a limited subset of random processes, ergodic processes in their turn are an even more limited subset of stationary processes.

The significance of the ergodic theorem is that it enables us to develop the statistics of the process $\zeta(t)$ using one, sufficiently long realisation. However, it is never possible to demonstrate ergodicity for ocean waves since experiments cannot be exactly repeated in the ocean as they can in the laboratory.

It can be shown that for a stationary wave process $\zeta(t)$ to be ergodic it is sufficient that its auto-correlation function $K(\tau)$ satisfies the following condition

$$K(\tau) = 0 \quad \text{at} \quad \tau \rightarrow \infty. \quad (1.77)$$

Let us now demonstrate the applicability of the ergodic theorem and condition (1.77) for a simple process. We assume that we have an ensemble of records of a process $\{\zeta_k(t)\} = z_k$. This means, that for a particular k , the process $\zeta_k(t)$ is constant and equal z_k . It is clear that process is stationary. For any chosen time t , any statistic, for example the mean, calculated across the ensemble gives the same number. However, when any single record $\zeta_{k=k_*}(t)$ is chosen at random and its time average calculated (Kinsman, 1965)

$$E[\zeta_{k=k_*}(t)]_t = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \zeta_{k=k_*}(t) dt, \quad (1.78)$$

then, it is obvious that

$$E[\zeta_k(t = t_*)]_k \neq E[\zeta_{k=k_*}(t)]_t. \quad (1.79)$$

Hence, this extremely simple process is stationary, but not ergodic. The condition (1.77) is clearly not satisfied as

$$K(\tau) = E[\zeta_{k=k_*}(t) \cdot \zeta_{k=k_*}(t + \tau)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T z_{k_*}^2 dt = z_{k_*}^2. \quad (1.80)$$

In the following we assume that the ergodic property holds for the stochastic process discussed in this text. Therefore, a single record $\zeta(t)$ will be used instead of an ensemble of records $\{\zeta_k(t)\}$.

The representation (1.28) is quite general but still a deterministic one. We will use it now for evaluation of the random wave model. We assume that $\zeta(t)$ is a stationary ergodic process satisfying the conditions (1.69) and (1.70). Without losing generality we can say that $\bar{\zeta} = 0$. Under the above assumptions there exists stochastic process $\tilde{A}(u, v, w)$, such that the process $\zeta(x, y, t)$ has the representation in form of the Fourier–Stieltjes integral⁸ (Yaglom, 1962)

$$\zeta(x, y, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp[i(ux + vy + wt)] d\tilde{A}(u, v, w). \quad (1.81)$$

A three-dimensional stochastic process $\tilde{A}(u, v, w)$ possesses uncorrelated increments which means that differences $[\tilde{A}(u'', v'', w'') - \tilde{A}(u', v', w')]$ for (u'', v'', w'') and (u', v', w') within a given cube in (u, v, w) space are uncorrelated with differences within any other non-overlapping cube.

The integrals (1.28) and (1.81) look similar, but they are significantly different. The former representation is a deterministic integral, while the latter is a representation of random function and should be defined by probabilistic limits.

The general representation given by Eq. (1.81) depends only on the assumption that process $\zeta(x, y, t)$ is ergodic and conditions (1.69) and (1.70) are valid. It also is important to note that this representation exists without assuming that the sea surface is composed of a superposition of linear waves.

When the surface displacements $\zeta(\mathbf{x}, t)$ is assumed to be a stationary random function of time and position, Eq. (1.81) can be represented as

$$\zeta(\mathbf{x}, t) = \int_{\mathbf{k}} \int_{\omega} \exp[i(\mathbf{k}\mathbf{x} - \omega t)] d\tilde{A}(\mathbf{k}, \omega), \quad (1.82)$$

⁸Fourier–Stieltjes integrals are a generalisation of both Fourier series and Fourier integrals. Applying the Fourier–Stieltjes integral we can represent functions which are not periodic and which are too large at infinity to be represented by ordinary Fourier integral.

in which

$$E[d\tilde{A}(k, \omega)d\tilde{A}^*(k', \omega')] = \begin{cases} 0 & \text{for } \mathbf{k}, \omega \neq \mathbf{k}', \omega' \\ \hat{\Psi}(\mathbf{k}, \omega)dkd\omega & \text{for } \mathbf{k} = \mathbf{k}', \omega = \omega'. \end{cases} \quad (1.83)$$

It should be noted that for stationary and homogeneous wave field, wavenumber-frequency spectrum $\hat{\Psi}(\mathbf{k}, \omega)$ is real and positive for all values of $\mathbf{k}$ and ω .

If the wave components obey the dispersion relation appropriate to infinitesimal gravity waves as

$$w^2 = g(u^2 + v^2)^{1/2} \tanh \left[(u^2 + v^2)^{1/2} h \right], \quad (1.84)$$

we should impose a restriction on wavenumber k and wave frequency ω . In particular, the parameters (u, v, w) can be replaced by the two parameters (ω, θ) , or alternatively by $\mathbf{k} = (u, v)$, and $d\tilde{A}(u, v, w)$ is zero whenever the above dispersion relation is not satisfied. Therefore Eq. (1.81) becomes the Fourier-Stieltjes integral representation in space (ω, θ) or (k_x, k_y)

$$\zeta(x, y, t) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] d\tilde{A}(\omega, \theta) \quad (1.85)$$

or

$$\zeta(x, y, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] dB(\mathbf{k}). \quad (1.86)$$

The models implied by Eqs. (1.85) and (1.86) permit both random phases and random amplitudes.

By definition, the cross-correlation function $K(X, Y, \tau)$ is

$$\begin{aligned} K(X, Y, \tau) &= E[\zeta(x, y, t) \cdot \zeta^*(x + X, y + Y, t + \tau)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} E\{\exp[i(ux + vy) - i\omega t] \\ &\quad \times \exp[-i[u'(x + X) + v'(y + Y)] + i\omega'(t + \tau)] \\ &\quad \times dA(\omega, \theta) dA^*(\omega', \theta')\}, \end{aligned} \quad (1.87)$$

in which $(*)$ denotes the conjugate of the complex variable.

The $dA(\omega, \theta) = d\tilde{A}(\omega, \theta)$ and $dA^*(\omega', \theta') = d\tilde{A}^*(\omega', \theta')$ and these are uncorrelated unless $u = u', v = v'$ and $\omega = \omega'$. Using the fact that variance of $\zeta(x, y, t)$ is finite, it can be shown that (Yaglom, 1962)

$$E[dA(\omega, \theta) dA^*(\omega', \theta')] = \hat{S}(\omega, \theta) \delta(\omega - \omega') \delta(\theta - \theta') d\omega d\omega' d\theta d\theta', \quad (1.88)$$

in which $\hat{S}(\omega, \theta)$ is a two-dimensional spectral density function of surface waves.

From Eq. (1.88) and the known relation for Dirac delta (Lighthill, 1975)

$$\int_{-\infty}^{\infty} f(y) \delta(x-y) dy = f(x), \quad (1.89)$$

Eq. (1.87) yields

$$\begin{aligned} K(X, Y, \tau) &= \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \exp[-i(uX + vY) + i\omega\tau] \hat{S}(\omega, \theta) d\omega d\theta \\ &= \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \exp[-i\mathbf{k} \cdot \mathbf{X} + i\omega\tau] \hat{S}(\omega, \theta) d\omega d\theta, \end{aligned} \quad (1.90)$$

where $\mathbf{X} = (X, Y)$.

In a similar way, from Eq. (1.87) we have

$$K(X, Y, \tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp[-i\mathbf{k} \cdot \mathbf{X} + i\omega\tau] \hat{\Psi}(\mathbf{k}) dk_x dk_y, \quad (1.91)$$

in which $\hat{\Psi}(\mathbf{k})$ is a wavenumber spectrum.

Let us now consider wave observation at a single point $P(x, y)$ where $X = Y = 0$. Expression (1.90) becomes the auto-correlation function

$$K(\tau) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \hat{S}(\omega, \theta) e^{i\omega\tau} d\omega d\theta \quad (1.92)$$

or

$$K(\tau) = \int_{-\infty}^{\infty} \tilde{S}(\omega) e^{i\omega\tau} d\omega, \quad (1.93)$$

in which

$$\tilde{S}(\omega) = \int_{-\pi}^{\pi} \hat{S}(\omega, \theta) d\theta. \quad (1.94)$$

The $\tilde{S}(\omega)$ is a frequency spectral density function or frequency spectrum of the ocean surface. It represents the distribution of wave energy in the frequency domain. Hence, the variance of surface waves is

$$E[\zeta^2] = \sigma_{\zeta}^2 = K(0) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \hat{S}(\omega, \theta) d\omega d\theta = \int_{-\infty}^{\infty} \tilde{S}(\omega) d\omega. \quad (1.95)$$

By taking the inverse Fourier transform, we have

$$\tilde{S}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\tau) e^{-i\omega\tau} d\tau. \quad (1.96)$$

The function $\tilde{S}(\omega)$ is called the symmetric function, as $\tilde{S}(\omega) = \tilde{S}(-\omega)$. This spectrum is usually used in theoretical considerations.

In experimental practise, the so called nonsymmetric spectrum $S(\omega)$ is used. It can be seen from Eqs. (1.93) and (1.96) that for ergodic stationary real random process $\zeta(t)$ with zero mean, the spectral density function $\tilde{S}(\omega)$ and the auto-correlation function $K(\tau)$ are a Fourier transform pair. This is called *the Wiener-Khinchine Theorem*.⁹ A few specific examples of auto-correlations $K(\tau)$ and spectral densities $\tilde{S}(\omega)$ are given in Tables 1.3 and 1.4.

For most processes, $K(\tau) \rightarrow \infty$. The decay time which characterise this behaviour is called correlation scale. For example, from Table 1.3 follows that a Markov process (see Chapter 4) has the auto-correlation function as

$$K(\tau) = K_0 \exp\left(-\frac{|\tau|}{\tau_0}\right). \quad (1.97)$$

For $\tau = \tau_0$ we have $K(\tau_0)/\tau_0 = 1/e \approx 0.368$. Taking the limit $\tau_0 \rightarrow 0$ in Eq. (1.97) gives the auto-correlation of a white noise random process (see Table 1.3).

The auto-correlation function $K(\tau)$ for surface elevation $\zeta(t)$ is a real and even function, i.e. $K(\tau) = K(-\tau)$. Therefore, the spectral density $\tilde{S}(\omega)$ is also a real and even function, and Eqs. (1.93) and (1.96) can be written as follows

$$K(\tau) = \int_{-\infty}^{\infty} \tilde{S}(\omega) e^{i\omega\tau} d\omega = 2 \int_0^{\infty} \tilde{S}(\omega) \cos(\omega\tau) d\omega \quad (1.98)$$

and

$$\tilde{S}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\tau) e^{-i\omega\tau} d\tau = \frac{1}{\pi} \int_0^{\infty} K(\tau) \cos(\omega\tau) d\tau. \quad (1.99)$$

We define nonsymmetric spectrum $S(\omega)$ as

$$S(\omega) = \begin{cases} 2\tilde{S}(\omega) & \text{for } \omega \geq 0 \\ 0 & \text{for } \omega < 0 \end{cases} \quad (1.100)$$

and

$$\tilde{S}(\omega) = \frac{1}{2} [S(\omega) + S(-\omega)]. \quad (1.101)$$

Therefore, Eqs. (1.98) and (1.99) take the forms frequently used in applications

$$K(\tau) = \int_0^{\infty} S(\omega) \cos(\omega\tau) d\omega \quad S(\omega) = \frac{2}{\pi} \int_0^{\infty} K(\tau) \cos(\omega\tau) d\tau. \quad (1.102)$$

⁹Norbert Wiener proved this theorem for the case of a deterministic function in 1930 and Aleksandr Khinchine formulated an analogous result for stationary stochastic process in 1934. The Wiener-Khinchine Theorem states that for a wide-sense process with an auto-correlation function $K(\tau)$ finite at every lag τ , there exists a monotonic function $\tilde{S}(\omega)$ in the frequency domain.

Table 1.3 Auto-correlation functions.

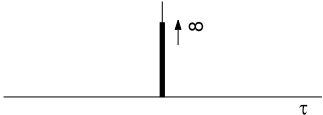
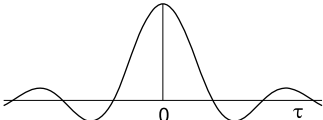
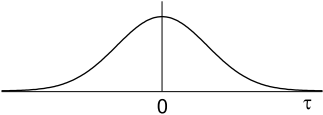
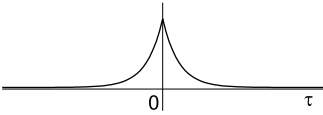
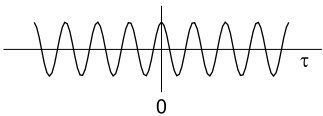
Process	$K(\tau) = \int_{-\infty}^{\infty} \tilde{S}(\omega) e^{i\omega\tau} d\omega$	
	Analytical expression	Graphical representation
White noise	$K_0 \delta(\tau)$	
Finite-band white noise	$K_0 \frac{\sin\left(\frac{\tau}{\tau_0}\right)}{\frac{\tau}{\tau_0}}$	
Gaussian	$K_0 \exp(-\alpha\tau^2)$	
Markov	$K_0 \exp\left(-\frac{ \tau }{\tau_0}\right)$	
Cosinusoidal signal with random phase	$K_0 \cos(\omega_0\tau)$	

Table 1.4 Spectral density functions.

$\tilde{S}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\tau) e^{-i\omega\tau} d\tau$	
Analytical expression	Graphical representation
$\frac{K_0}{\pi}$	
$\begin{matrix} K_0 \tau_0 & \text{for } \omega \leq \tau_0^{-1} \\ 0 & \text{for } \omega > \tau_0^{-1} \end{matrix}$	
$\frac{K_0}{\sqrt{\pi\alpha}} \exp\left(-\frac{\omega^2}{4\alpha}\right)$	
$\frac{2K_0}{\pi} \frac{\tau_0}{1 + (\omega\tau_0)^2}$	
$\frac{K_0}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	

Let us consider for a simple harmonic wave $\zeta(x, t) = a \cos(kx - \omega t)$ with $\omega = \omega_0$ (see Tables 1.3 and 1.4). Thus we obtain

$$K(\tau) = K_0 \cos(\omega_0 \tau), \quad \tilde{S}(\omega) = \frac{K_0}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]. \quad (1.103)$$

For $\omega_0 > 0$ we have

$$S(\omega) = K_0 \delta(\omega - \omega_0) \quad \text{with} \quad K_0 = \frac{a^2}{2} = \overline{\zeta^2}. \quad (1.104)$$

Let us consider now the wavenumber spectrum $\hat{\Psi}(\mathbf{k})$ for $X = Y = 0$. From Eq. (1.91) we have

$$K(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i\omega\tau} \hat{\Psi}(\mathbf{k}) \, dk_x \, dk_y = \int_0^{\infty} \int_{-\pi}^{\pi} e^{i\omega\tau} \hat{\Psi}(k, \theta) \, k \, dk \, d\theta, \quad (1.105)$$

in which $k = |\mathbf{k}|$.

In deep water, the dispersion relation (1.17) becomes $\omega^2 = gk$ and $k \, dk = 2 \frac{\omega^3}{g^2} d\omega$. After substituting this into Eq. (1.105) we obtain

$$K(\tau) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} e^{i\omega\tau} \left\{ \frac{2\omega^3}{g^2} \hat{\Psi}(k, \theta) \Big|_{k=\omega^2/g} \right\} d\omega d\theta. \quad (1.106)$$

Now Eq. (1.106) can be compared with Eq. (1.92), i.e.

$$\hat{S}(\omega, \theta) = \frac{2\omega^3}{g^2} \hat{\Psi}(k, \theta) \quad (1.107)$$

and

$$S(\omega) = \frac{2\omega^3}{g^2} \int_{-\pi}^{\pi} \hat{\Psi}(k, \theta) \, d\theta. \quad (1.108)$$

Thus, the frequency spectrum $S(\omega)$ represents an integral over the wavenumber spectrum $\hat{\Psi}(k, \theta)$ for all directions θ , when k is constant and equals ω^2/g .

The above definitions can be immediately extended to a set of two random processes, simultaneously recorded at the same point or at two different points. As an example, we will first consider the surface elevation $\zeta(t)$ and the horizontal component of orbital velocity $u(t)$ recorded at point $P(0, 0)$. Using the linear wave theory we express velocity $u(t)$ as corresponding to surface elevation $\zeta(t)$, given by Eq. (1.85), i.e.

$$u(0, 0, z, t) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \frac{gk \cos \theta}{\omega} \frac{\cosh k(z+h)}{\cosh kh} \exp(-i\omega t) d\tilde{A}(\omega, \theta), \quad (1.109)$$

in which z is a level at which velocity u is recorded.

The cross-correlation function $K_{\zeta u}(\tau)$ takes the form

$$\begin{aligned} K_{\zeta u}(\tau, z) = E[\zeta(t) u(t + \tau)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\pi}^{\infty} \int_{-\pi}^{\infty} \frac{gk' \cos \theta'}{\omega'} \\ &\times \frac{\cosh k(z + h)}{\cosh kh} E\{\exp[i(\omega' - \omega) + i\omega'\tau]\} \\ &\times E\{dA(\omega, \theta) dA^*(\omega', \theta')\} \end{aligned} \quad (1.110)$$

or

$$K_{\zeta u}(\tau, z) = \int_{-\infty}^{\infty} \left\{ \int_{-\pi}^{\pi} \frac{gk \cos \theta}{\omega} \frac{\cosh k(z + h)}{\cosh kh} \hat{S}(\omega, \theta) d\theta \right\} e^{i\omega\tau} d\omega. \quad (1.111)$$

Defining $S_{\zeta u}(\omega, z)$ as

$$S_{\zeta u}(\omega, z) = \int_{-\pi}^{\pi} \frac{gk \cos \theta}{\omega} \frac{\cosh k(z + h)}{\cosh kh} \hat{S}(\omega, \theta) d\theta, \quad (1.112)$$

the Wiener–Khinchine Theorem yields

$$K_{\zeta u}(\tau, z) = \int_{-\infty}^{\infty} S_{\zeta u}(\omega, z) e^{i\omega\tau} d\omega \quad (1.113)$$

and

$$S_{\zeta u}(\omega, z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} K_{\zeta u}(\tau, z) e^{-i\omega\tau} d\tau. \quad (1.114)$$

In general $S_{\zeta u}(\omega, z)$ is a complex function, i.e.

$$S_{\zeta u}(\omega, z) = C_{\zeta u}(\omega, z) + i Q_{\zeta u}(\omega, z), \quad (1.115)$$

in which the real part $C_{\zeta u}(\omega, z)$ is referred to as the *co-spectrum*, while the imaginary part $Q_{\zeta u}(\omega, z)$ is referred to as the *quadrature spectrum*. The amplitude spectrum of $S_{\zeta u}(\omega, z)$ becomes

$$|S_{\zeta u}(\omega, z)| = \sqrt{[C_{\zeta u}(\omega, z)]^2 + [Q_{\zeta u}(\omega, z)]^2}, \quad (1.116)$$

and the phase spectrum is given by

$$\varphi(\omega, z) = \tan^{-1} \left\{ \frac{Q_{\zeta u}(\omega, z)}{C_{\zeta u}(\omega, z)} \right\}^{1/2}. \quad (1.117)$$

The various properties of the co- and quadrature spectra can be found elsewhere (for example, Bendat and Piersol, 1986; Ochi, 1998).

It should be noted that the two-dimensional spectrum of surface waves $\hat{S}(\omega, \theta)$ is commonly represented as a product of a frequency spectrum $S(\omega)$ and directional spreading function $D(\theta, \omega)$, i.e.

$$\hat{S}(\omega, \theta) = S(\omega) D(\theta, \omega). \quad (1.118)$$

Function $D(\theta, \omega)$ will be discussed in detail in Section 3.4.

After substituting Eq. (1.118) into Eq. (1.112) and using Eq. (1.115), we obtain

$$\left. \begin{aligned} C_{\zeta u}(\omega, z) &= \frac{gk}{\omega} \frac{\cosh k(z+h)}{\cosh kh} S(\omega) \int_{-\pi}^{\pi} D(\theta; \omega) \cos \theta d\theta \\ Q_{\zeta u}(\omega, z) &= 0 \end{aligned} \right\}. \quad (1.119)$$

The vanishing of the quadrature spectrum $Q_{\zeta u}(\omega, z)$ is a consequence of the assumption that sea surface oscillation $\zeta(t)$ and wave-induced horizontal velocity $u(t)$ are given in the same point and that there is no time delay between them. This conclusion is valid for the symmetrical directional spreading function, i.e. when $D(\theta) = D(-\theta)$.

We now consider the case when surface elevation is simultaneously recorded in two points, i.e. point $P(x, y,)$ and $P_1(x+X, y+Y)$. Eq. (1.90) for cross-correlation function yields

$$K(X, Y, \tau) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} \exp \{-ikd \cos(\theta - \theta_0) + i\omega\tau\} \hat{S}(\omega, \theta) d\omega d\theta, \quad (1.120)$$

in which $d = \sqrt{X^2 + Y^2}$ and $\arctan \theta_0 = Y/X$. Now the Wiener-Khinchine Theorem gives

$$S(\omega; d, \theta_0) = \int_{-\pi}^{\pi} \exp[-ikd \cos(\theta - \theta_0)] \hat{S}(\omega, \theta) d\theta. \quad (1.121)$$

Using the identity (Abramowitz and Stegun, 1975)

$$\exp[ikd \cos(\theta - \theta_0)] = \sum_{m=0}^{\infty} \epsilon_m i^m J_m(kd) \cos m(\theta - \theta_0), \quad (1.122)$$

in which $\epsilon_0 = 1$ and $\epsilon_m = 2$ ($m \geq 1$), $J_m(x)$ is a first kind of Bessel function of m order and separating the real and imaginary parts we obtain co-spectrum $C(\omega; d, \theta_0)$ and quadrature spectrum $Q(\omega; d, \theta_0)$ in the form

$$C(\omega; d, \theta_0) = J_0(kd) + 2 \sum_{m=1}^{\infty} (-1)^m J_{2m}(kd) \cos[2m(\theta - \theta_0)] \quad (1.123)$$

and

$$-Q(\omega; d, \theta_0) = 2 \sum_{m=0}^{\infty} (-1)^m J_{2m+1}(kd) \cos[(2m+1)(\theta - \theta_0)]. \quad (1.124)$$

1.4.3 Wave energy balance in spectral form

Wave generation models provide the linkage between the wavenumber-frequency spectrum of air pressure fluctuations $\hat{\Psi}_a(\mathbf{k}, \omega)$ and the wavenumber spectrum of surface oscillations $\hat{\Psi}(\mathbf{k}, \mathbf{x}, t)$ in a simpler case when waves propagate in deep water or in water of finite but constant depth. Generally, when water depth changes in space, and island or artificial obstacles are present, the wind generated waves will undergo several other processes such as refraction, diffraction and dissipation due to bottom friction, whitecapping and breaking in the surf zone.

Taking into account the random character of wave motion, it seems that the most appropriate method of evaluation of waves, in time and space are the spectral methods. Resulting functions of these methods, i.e. the spectral densities $\hat{S}(\omega, \theta)$ or $\hat{\Psi}(\mathbf{k})$, provide the distribution of wave energy in the frequency or wavenumber space.

In order to develop an energy balance equation we assume for a moment that the dispersion relation does not depend on time, but is rather a slowly changing function of space coordinates. It can also depend on local properties such as water depth, current velocity or ambient density current, i.e.

$$\omega = \Omega[\mathbf{k}, f(\mathbf{x}, h, \mathbf{u}, \varrho, \dots)]. \quad (1.125)$$

If the medium itself is moving with velocity $\mathbf{u}$, the frequency of waves passing a field point is

$$\omega = \Omega(\mathbf{k}, \mathbf{x}) = \sigma + \mathbf{k} \cdot \mathbf{u} = [gk \tanh(kh)]^{1/2} + \mathbf{k} \cdot \mathbf{u}. \quad (1.126)$$

Usually the quantity ω is called the observed or apparent frequency, while σ is the intrinsic frequency whose functional dependence on $\mathbf{k}$ is known as a classical dispersion relation

$$\sigma^2 = gk \tanh(kh). \quad (1.127)$$

It is well known that the number of wave crests passing a given point per unit time is determined by the kinematics conservation equations (Massel, 1989)

$$\frac{\partial \mathbf{k}}{\partial t} + \nabla_h \omega = 0, \quad \frac{\partial k_i}{\partial x_j} - \frac{\partial k_j}{\partial x_i} = 0, \quad i, j = 1, 2. \quad (1.128)$$

Substituting Eq. (1.125) into Eq. (1.128) gives

$$\frac{\partial k_i}{\partial t} + C_{g_j}(\mathbf{x}, \mathbf{k}, t) \frac{\partial k_j}{\partial x_i} = - \frac{\partial \Omega}{\partial x_i}, \quad (1.129)$$

in which $C_{g_j} = \frac{\partial \Omega}{\partial k_j}$ is a group velocity. Eq. (1.129) can be rewritten as

$$\frac{dk_i}{dt} = -\frac{\partial \Omega}{\partial x_i} \quad \text{and} \quad \frac{dx_i}{dt} = \frac{\partial \Omega}{\partial k_i}. \quad (1.130)$$

Equations (1.130) are identical to the Hamilton equations in mechanics. The vectors $\mathbf{x}$ should be interpreted as coordinates and vectors $\mathbf{k}$ represent the moments, and the frequency $\Omega(\mathbf{k}, \mathbf{x}, t)$ is taken to be the Hamiltonian (Whitham, 1974).

For free surface waves, the mean densities of potential and kinetic energy are equal. Therefore, $2\hat{\Psi}(\mathbf{k}, \mathbf{x}, t)$ is a mean spectral density of total wave energy in the unit volume $d\mathbf{k} \cdot d\mathbf{x}$ in phase space $(\mathbf{k}, \mathbf{x})$ at a given time t . According to Liouville's theorem, this energy should be conservative during volume evolution

$$\frac{d\hat{\Psi}}{dt} = \frac{\partial \hat{\Psi}}{\partial t} + \frac{\partial x_i}{\partial t} \frac{\partial \hat{\Psi}}{\partial x_i} + \frac{\partial k_i}{\partial x_i} \frac{\partial \hat{\Psi}}{\partial k_i} = 0 \quad (1.131)$$

or

$$\frac{\partial \hat{\Psi}}{\partial t} + \frac{\partial \Omega}{\partial k_i} \frac{\partial \hat{\Psi}}{\partial x_i} - \frac{\partial \Omega}{\partial x_i} \frac{\partial \hat{\Psi}}{\partial k_i} = 0, \quad (1.132)$$

when Eq. (1.130) is used.

If the wave field is subjected to processes of generation, dissipation, nonlinear interaction between spectral components and other possible interactions with atmospheric boundary layer and various ocean movements (currents, internal waves, etc.), Eq. (1.132) should be supplemented by a source-sink term at the right-hand side, i.e.

$$\frac{\partial \hat{\Psi}}{\partial t} + \frac{\partial \Omega}{\partial k_i} \frac{\partial \hat{\Psi}}{\partial x_i} - \frac{\partial \Omega}{\partial x_i} \frac{\partial \hat{\Psi}}{\partial k_i} = Q(\mathbf{k}, \mathbf{x}, t), \quad (1.133)$$

in which $Q = \sum_i Q_i$.

The first term at the left-hand side of Eq. (1.133) expresses the local evolution of the spectrum in time, while the second term represents the evolution of the spectrum for the horizontally non-homogeneous wave field. This term shows that energy is transported at the group velocity. The third term reflects the effects of refraction and shoaling due to a non-horizontal bottom or due to current. The spectral energy balance equation is also known as the radiative transfer equation or the transport equation. In particular, an analogy between the sea and a kinetic gas, which explains the name kinetic equation, is used by Russian oceanographers.

In engineering and oceanographic practise the two-dimensional frequency-directional spectrum $\hat{S}(\omega, \theta; \mathbf{x}, t)$ is frequently used, rather than

the wavenumber spectrum $\hat{\Psi}(\mathbf{k}; \mathbf{x}, t)$ (see Chapter 3). To evaluate spectrum $\hat{S}(\omega, \theta; \mathbf{x}, t)$, we adopt the following variables transformation

$$k_1 = \chi(\omega, \mathbf{x}) \cos \theta, \quad k_2 = \chi(\omega, \mathbf{x}) \sin \theta, \quad (1.134)$$

where function $\chi(\omega, \mathbf{x}) = k$ is a solution of the dispersion relation.

Therefore variables transformation (1.134) now becomes

$$\hat{\Psi}(\mathbf{k}; \mathbf{x}, t) = J \hat{S}(\omega, \theta; \mathbf{x}, t) = f(\omega, \theta; \mathbf{x}, t), \quad (1.135)$$

in which J is a Jacobian of the transformation, i.e.

$$J = \frac{\partial(\omega, \theta)}{\partial(k_1, k_2)} = \left[\frac{\partial(k_1, k_2)}{\partial(\omega, \theta)} \right]^{-1}. \quad (1.136)$$

Thus

$$J = \left\{ \begin{vmatrix} \frac{\partial \chi}{\partial \omega} \cos \theta & -\chi \sin \theta \\ \frac{\partial \chi}{\partial \omega} \sin \theta & -\chi \cos \theta \end{vmatrix} \right\}^{-1} = \frac{C_g}{\chi}. \quad (1.137)$$

After substituting Eq. (1.135) into Eq. (1.133) we obtain (Krasitskii, 1974)

$$\begin{aligned} \frac{\partial f}{\partial t} + C_g \cos \theta \frac{\partial f}{\partial x} + C_g \sin \theta \frac{\partial f}{\partial y} + C_g \left(\cos \theta \frac{\partial \ln \chi}{\partial y} - \sin \theta \frac{\partial \ln \chi}{\partial x} \right) \frac{\partial f}{\partial \theta} \\ = \frac{C_g}{\chi} Q(\omega, \theta, \mathbf{x}, t). \end{aligned} \quad (1.138)$$

As $C_p = \omega/\chi = \omega/k$, we can rewrite Eq. (1.138) as follows

$$\begin{aligned} \frac{\partial}{\partial t}(C_p C_g \hat{S}) + C_g \cos \theta \frac{\partial}{\partial x}(C_p C_g \hat{S}) + C_g \sin \theta \frac{\partial}{\partial y}(C_p C_g \hat{S}) \\ + \frac{C_g}{C_p} \left[\sin \theta \frac{\partial C_p}{\partial x} - \cos \theta \frac{\partial C_p}{\partial y} \right] \frac{\partial}{\partial \theta}(C_p C_g \hat{S}) \\ = C_p C_g Q(\omega, \theta, \mathbf{x}, t), \end{aligned} \quad (1.139)$$

in which $Q = \sum_i Q_i$. Some solutions of Eq. (1.138) (or (1.139)) will be discussed in Chapters 6 and 7.

Equations (1.138) and (1.139) can be applied for a dispersion relation not depending on time, when an ambient current is uniform in space. If waves propagate on a non-stationary and non-uniform current $\mathbf{u}(\mathbf{x}, t)$, the intrinsic frequency σ may vary in space and time. In such a case, the

spectral energy density will not be conserved. However Bretherton and Garrett (1969) showed that the quantity called wave action density defined as $N = \hat{\Psi}(\mathbf{k}; \mathbf{x}, t)/\sigma$ is conserved in a moving medium. Therefore, instead of Eq. (1.133), we apply a more general principle of conservation of wave action density N in the form

$$\frac{\partial N}{\partial t} + \frac{\partial \Omega}{\partial k_i} \frac{\partial N}{\partial x_i} - \frac{\partial \Omega}{\partial x_i} \frac{\partial N}{\partial k_i} = \frac{Q}{\sigma}. \quad (1.140)$$

In order to illustrate the relationship between wave action density N and energy density $\hat{\Psi}$ let us consider the simplest example of a steady unidirectional wave train in deep water on a variable current $u(x)$ (source term $Q \equiv 0$). Then, Eq. (1.140) reduces to

$$\frac{\partial}{\partial x} \left[\hat{\Psi} \left(\frac{u + C_g}{\sigma} \right) \right] = 0. \quad (1.141)$$

For gravity waves in deep water $\sigma = g/C_p$ and $C_g = 1/2 C_p$. Thus

$$\hat{\Psi} \left(u + \frac{1}{2} C_p \right) C_p = \text{const.} \quad (1.142)$$

Although the flux of wave action is constant, the flux of wave energy is not. For example, as C_p decreases in an adverse current, the energy flux of the wave motion increases because of the work done by ‘radiation stress’ (Longuet-Higgins and Stewart, 1964).

The basic difficulty in solving Eq. (1.133) or (1.140) is an evaluation of the function Q . In general, Q is a function of wavenumber $\mathbf{k}$ and a functional of $\hat{\Psi}$, i.e. $Q = Q[\mathbf{k}, \hat{\Psi}]$. Function Q can also be dependent on other parameters which are functions of $\mathbf{x}$ and t (for example, wind velocity).

The theory of nonlinear weak interactions for wind-induced waves was first formulated by Hasselmann (1962, 1968). Assuming that

$$\frac{1}{k} \frac{\partial \hat{\Psi}}{\partial k_i} \ll 1 \quad \text{and} \quad \frac{1}{\omega} \frac{\partial \hat{\Psi}}{\partial t} \ll 1, \quad (1.143)$$

with the accuracy of the second approximation in interaction equations, Hasselmann evaluated the general form of source/sink function Q as follows (Hasselmann, 1968; Willebrand, 1975)

$$Q(\mathbf{k}) = \sum_{i=1}^{i=9} Q_i, \quad (1.144)$$

in which

$$Q_1 = \alpha, \quad (1.145)$$

$$Q_2 = \beta \hat{\Psi}(\mathbf{k}), \quad (1.146)$$

$$Q_3 = \hat{\Psi}(\mathbf{k}) \int \hat{\gamma}(\mathbf{k}, \mathbf{k}') \hat{\Psi}(\mathbf{k}') d\mathbf{k}', \quad (1.147)$$

$$Q_4 = -\delta \hat{\Psi}(\mathbf{k}) + \int \epsilon(\mathbf{k}, \mathbf{k}') \hat{\Psi}(\mathbf{k}') d\mathbf{k}', \quad (1.148)$$

$$Q_5 = \int \left[T_1 \hat{\Psi}(\mathbf{k}') \hat{\Psi}(\mathbf{k}'') \hat{\Psi}(\mathbf{k} - \mathbf{k}' - \mathbf{k}'') \right. \\ \left. - T_2 \hat{\Psi}(\mathbf{k}) \hat{\Psi}(\mathbf{k}') \hat{\Psi}(\mathbf{k}'') \right] d\mathbf{k}' d\mathbf{k}'', \quad (1.149)$$

$$Q_6 = -\nu_{ij} k_i k_j \hat{\Psi}(\mathbf{k}), \quad (1.150)$$

$$Q_7 = -\hat{\gamma} \hat{\Psi}(\mathbf{k}). \quad (1.151)$$

The coefficients $\alpha, \beta, \delta, \epsilon, \hat{\gamma}$ depend on the properties of the interacting fields. Similar expressions for source/sink function Q in Eq. (1.139) are obtained after appropriate variables transformation.

The terms Q_1 and Q_2 determine the wind input which may be added or subtracted from the momentum of existing waves. In particular, term Q_1 describes energy flux due to turbulent fluctuations of surface pressure (Phillips' mechanism). Term Q_2 corresponds to Miles' coupling mechanism between the wave field and the mean boundary layer flow. The term Q_3 is a nonlinear correction of Miles' theory and Q_4 is responsible for interaction of the wave motion with atmospheric turbulence. The term Q_5 expresses nonlinear interaction between spectral components when energy is moved from one part of the spectrum to another. In general, nonlinear interactions in the wave field are weak, as the evolution time of the wave field is much greater than the characteristic wave period and the evolution in space is much greater than the characteristic wavelength. Therefore, the process of resonant wave-wave interaction may be conservative with no net loss or gain of energy over the spectrum. However, in some regions and at some times, the interactions may be quite strong. For example, strong interactions are induced by wave breaking.

The term Q_6 represents the energy changes due to interaction of waves with the mean current, while Q_7 expresses the energy changes due to interaction of waves with turbulence in the water layer. The two other components of source function Q (Q_8 and Q_9) describe the strong nonlinearities. The term Q_8 is related to energy dissipation due to bottom friction and Q_9 represents the dissipation of wave energy due to wave breaking. Wave breaking is a strongly nonlinear interaction in which wave energy is moving from breaking waves to turbulence and to waves longer than waves corresponding to the spectrum peak.

The most important components of function Q are the terms which describe the waves generation by wind: $Q_{\text{in}} = Q_1 + Q_2$, the nonlinear interaction between wave components: $Q_{\text{nl}} = Q_5$ and energy dissipation due to bottom friction and due to wave breaking: $Q_{\text{diss}} = Q_8 + Q_9$. Further discussion of the particular source Q_i components, their importance for wave prediction models and their last improvements is given in Chapters 6 and 7.

1.4.4 Representation of wave properties in time-frequency space

1.4.4.1 Wavelet transform approach

Most of the oceanographic signals are the time-domain signals. However, in many cases the most distinguished information is hidden in the frequency spectrum which provides the energy associated with a given frequency. The frequency spectrum of the signal can be obtained by the Fourier transform. The Fourier transform yields an information on how much but not when (in time) the particular frequency components exist. Such information is sufficient in a case of the stationary signals as the frequency content of such signals does not change in time and all frequency components exist all the time. When the time localisation of the spectral components is required, the transform of time series which provides the time-frequency representation of the signal should be developed. A transform of such type is the Wavelet transform which gives full time-frequency representation of the time series (Torrence and Compo, 1998). In general, the Wavelet transform (WT) of the signal, $x(t)$, is defined as a following inner product

$$WT(\tau, b) = \langle g_{\tau b} | x \rangle = \int_{-\infty}^{\infty} x(t) g_{\tau b}^*(t; \tau, b) dt. \quad (1.152)$$

The family of continuously translated and dilated wavelets is generated from mother wavelet $g(t)$

$$g_{\tau b}(t; \tau, b) = \frac{1}{\sqrt{b}} g\left(\frac{t - \tau}{b}\right), \quad (1.153)$$

where τ is the translation parameter, corresponding to the position of the wavelet as it is shifted through the signal, b is the scale dilation parameter determining the width of the wavelet and the $(*)$ indicates the complex conjugate. The scale $b > 1$ dilates (or stretches out) the signals, whereas scale $b < 1$ compresses the signal. The wavelet coefficients, $WT(\tau, b)$, represent the correlation (in term of the time-scale functions) between the wavelet and a localised section of the signal. If the signal has a major component of the frequency corresponding to the given scale, then the wavelet at this scale is close to the signal at the particular location and the corresponding wavelet transform coefficient, determined at this point, has a relatively large value. Therefore, the Wavelet transform is a sort of a *microscope* with magnification $1/b$ and location given by parameter τ , while the optics of the microscope is characterised by the function $g_{\tau b}(t; \tau, b)$. For the wavelet which has the mother wavelet status, the function $g(t)$ must be regular function such that it decays to zero in the limit $|t| \rightarrow \infty$ and it has zero mean.

The Wavelet transform should reflect the type of features which are present in the time series. For time series with sharp steps, a boxcar-like wavelet should be chosen, while for smoothly varying time series a smooth function is more appropriate. However, if the wavelet power spectra are not of the primary interest, the choice of wavelet function is not critical. One of the most extensively used mother wavelet is the Morlet's wavelet

$$g(t) = \exp\left(-\frac{1}{2} t^2\right) \exp(ict). \quad (1.154)$$

Eq. (1.154) represents a plane wave of frequency c , modulated by a Gaussian envelope of the unit width. In the oceanographic applications, the other mother wavelets, such as the orthogonal wavelets (Yamada and Ohkitani, 1990; Mori and Yasuda, 1994), Paul's wavelet or DOG wavelet (derivative of a Gaussian) (Torrence and Compo, 1998) are also used. Using the representation (1.153), the Morlet wavelet takes the form (Massel, 2001b)

$$g_{\tau b}(t) = \frac{1}{\sqrt{b}} \exp\left[-\frac{1}{2} \left(\frac{t - \tau}{b}\right)^2\right] \exp\left[ic \frac{t - \tau}{b}\right]. \quad (1.155)$$

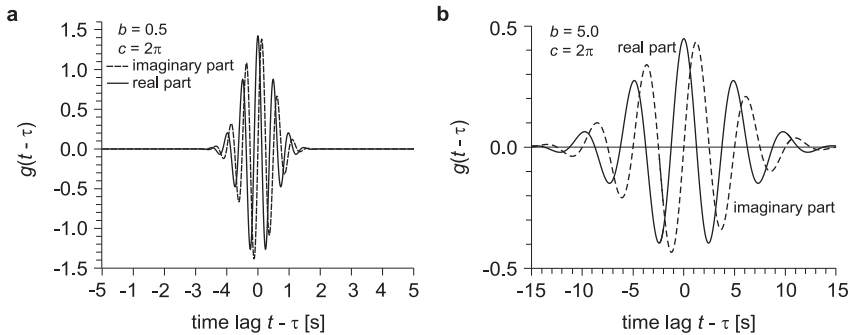


Fig. 1.8 The Morlet's wavelet Eq. (1.155) of a constant value of c and different scales b (a) $b = 0.5$, $c = 2\pi$; (b) $b = 5.0$, $c = 2\pi$ (adapted from Massel, 2001b).

The frequency nature of the parameter c is clearly seen when we take $c = 2\pi$. The Eq. (1.155) becomes

$$g_{\tau b}(t) = \frac{1}{\sqrt{b}} \exp \left[-\frac{1}{2} \left(\frac{t - \tau}{b} \right)^2 \right] \exp \left[i \frac{2\pi}{b} (t - \tau) \right]. \quad (1.156)$$

The term $\exp \left[i \frac{2\pi}{b} (t - \tau) \right]$ represents now the plane sinusoidal wave with a frequency $2\pi/b$ or with the scale dilation b which can be treated as a period.

The real and imaginary parts of the wavelet (1.156) for various scale parameters b are illustrated in Fig. 1.8. As the b increases, a width of the Gaussian spreads in time from its centre value. If the scale parameter b is constant and frequency parameter c increases, the number of oscillations over the span of the function also increases. This effect is clearly seen by comparison of Fig. 1.8 and Fig. 1.9.

The wavelet analysis is done in a such a way that the initial signal is multiplied with a window (the wavelet). The window width is changed as the transform is computed for every single spectral component. This is the most significant characteristic of the Wavelet transform. Changing the window width influences the resolution of the transform. It is schematically illustrated schematically in Fig. 1.10. Every box corresponds to the value of the Wavelet transform in the translation-scale plane. Although the widths and heights of the boxes change, their areas are constant (for example, see two hatched boxes). Each box represents the same portion of the time-frequency plane, but gives different ratios of time and frequency. At low frequency (high value of scale b), the height of the box is small, but its width

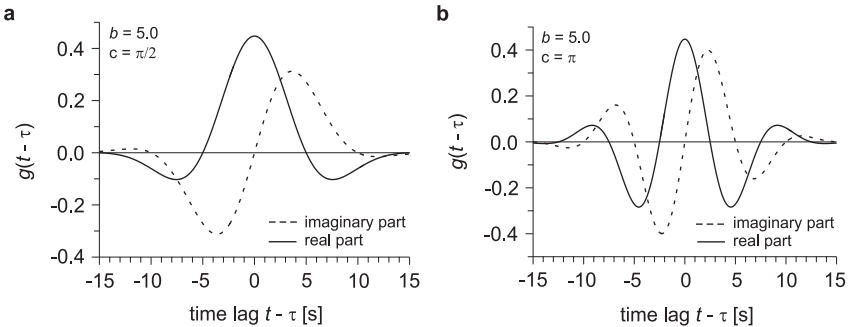


Fig. 1.9 The Morlet's wavelet Eq. (1.155) of a constant scale b and different c parameters (a) $b = 5.0$, $c = \pi/2$; (b) $b = 5.0$, $c = \pi$ (adapted from Massel, 2001b).

is large, i.e. there is better frequency resolution and poor time resolution. On the other hand, at higher frequencies (low scale b), the height of the boxes increases and the width of the boxes decreases. This corresponds to a very narrow window when the frequency resolution is going poorer and the time resolution is going better.

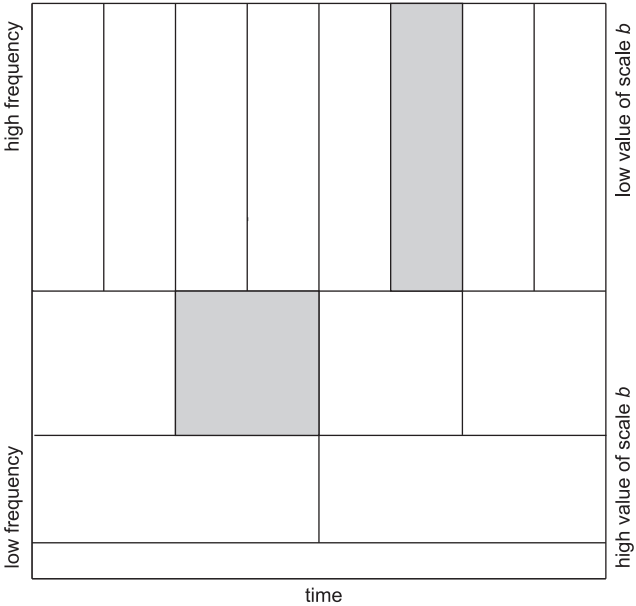


Fig. 1.10 Resolution scheme in the time-frequency plane (adapted from Massel, 2001b).

Now we will examine some energy properties of the wavelet. Firstly, the Wavelet transform conserves the energy, i.e.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = C^{-1} \int_0^{\infty} \int_0^{\infty} |WT(\tau, b)|^2 b^{-2} d\tau db, \quad (1.157)$$

in which

$$C^{-1} = \int_{-\infty}^{\infty} \left(\omega^{-1} |G(\omega)|^2 \right) d\omega, \quad (1.158)$$

and the $G(\omega)$ is the Fourier transform of function $g(t)$.

Using the Wavelet transform and the coefficient C we can define various wavelet energy spectra and spectral densities. In particular, so called *time-scale energy density* becomes

$$E_1(\tau, b) = \frac{|WT(\tau, b)|^2}{b}. \quad (1.159)$$

By integrating Eq. (1.159) versus scale b we obtain the *local energy density* (Farge, 1992)

$$E_2(\tau) = C^{-1} \int_0^{\infty} E_1(\tau, b) \frac{db}{b}. \quad (1.160)$$

On the other hand, the integration of Eq. (1.159) versus time τ gives the *global wavelet energy spectrum* $E_3(b)$ as

$$E_3(b) = \int_0^{\infty} E_1(\tau, b) d\tau. \quad (1.161)$$

Torrence and Compo (1998) noted that the smoothed Fourier spectrum approaches the global wavelet spectrum more closely when the amount of necessary smoothing decreasing with increasing scale.

Finally, the *total energy* of the time series $x(t)$ becomes

$$E = C^{-1} \int_0^{\infty} E_3(b) \frac{db}{b}. \quad (1.162)$$

After substituting Eqs. (1.159) and (1.161) into Eq. (1.162) we get

$$E = C^{-1} \int_0^{\infty} \int_{-\infty}^{\infty} E_1(\tau, b) \frac{d\tau db}{b} = C^{-1} \int_0^{\infty} \int_{-\infty}^{\infty} |WT(\tau, b)|^2 \frac{d\tau db}{b^2}, \quad (1.163)$$

what constitutes the conservation of energy (1.157).

In Figs. 1.11 and 1.12, the contours of $|WT|$ and global wavelet energy spectrum E_3 of the “chirp” signal are given. The frequency components,

contributed to the signal, and time limits are as follows (Massel, 2001b)

$$\zeta(t) = \begin{cases} 1.0 \cdot \sin\left(\frac{2\pi t}{2}\right) & \text{for } 5 \text{ s} < t < 20 \text{ s}, \\ 1.0 \cdot \sin\left(\frac{2\pi t}{5}\right) & \text{for } 20 \text{ s} < t < 40 \text{ s}, \\ 1.0 \cdot \sin\left(\frac{2\pi t}{10}\right) & \text{for } 40 \text{ s} < t < 50 \text{ s}. \end{cases} \tag{1.164}$$

Maximum values of the $|WT|$ correspond to the scale parameters equivalent to the periods of signal components. However, the maximum values of $|WT|$ are shifting to the higher scales as the periods of the initial signal components increase. Fig. 1.11 demonstrates that Wavelet transform is capable to provide the time and frequency information simultaneously, giving a full time-frequency representation of the signal. In contrast to the Fourier transform, the Wavelet transform allows exceptional localisation both, in the time and in the scale (frequency) domains.

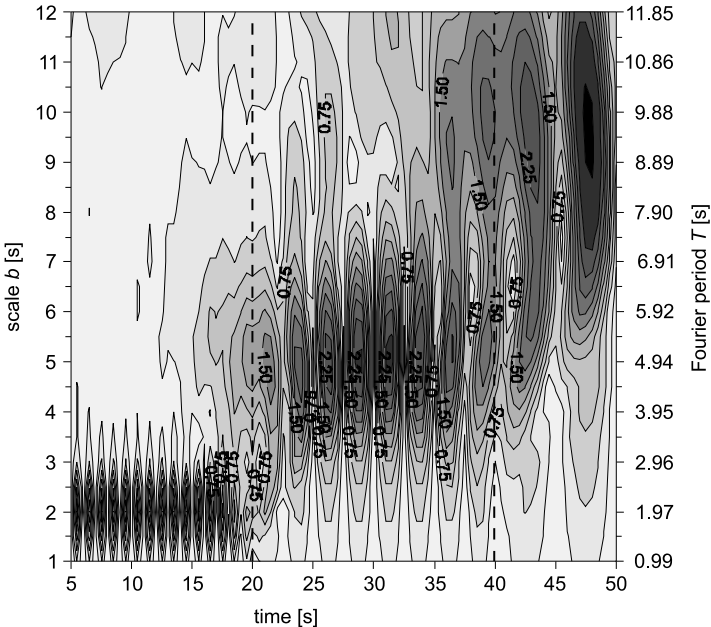


Fig. 1.11 Contours of the $|WT|$ for the “chirp” type signal (adapted from Massel, 2001b).

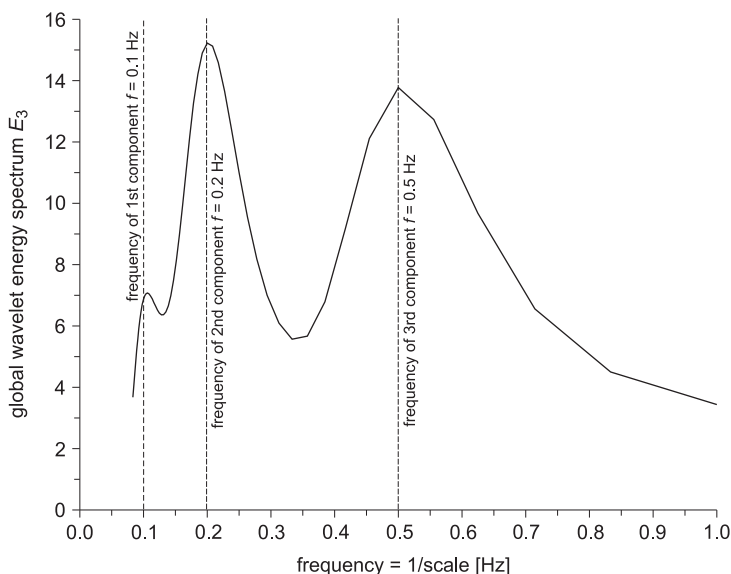


Fig. 1.12 Global wavelet energy spectrum E_3 for “chirp” signal (adapted from Massel, 2001b).

The global wavelet spectrum $E_3(b)$ in Fig. 1.12 is plotted versus the quantity $f = 1/b$ which can be treated as a measure of frequency. The dominant contribution into the global wavelet spectrum is provided by the signal components with frequencies $f = 0.1$ Hz, 0.2 Hz and 0.5 Hz. The spectrum E_3 mimics the commonly used frequency spectrum. However it should be pointed out that there is not a straightforward correspondence between E_3 and frequency spectrum based on the Fourier transform.

Wavelets have been successfully implemented in signal and image processing, ordinary and partial differential equations theory, numerical analysis and communication theory. On the other hand, the application of the Wavelet transform to the fields of ocean engineering and oceanography is not frequent. Meyers *et al.* (1993) demonstrated the usefulness of the wavelet transform in studying dispersion of Yanai waves in a reduced gravity equatorial model. In a paper by Shen and Mei (1994), the continuous Wavelet transform was developed to analyse the energy balance in the equilibrium spectral sub-range of wind-generated gravity waves. Scott (2005) estimated the statistics of steep waves using the wavelet methodology to obtain wave gauge data. Mori and Yasuda (1994), and Liu and Babanin (2004) applied the Wavelet transform to detect wave growth and breaking

in the time series. A sudden surface jump associated with breaking wave is regarded as a shock wave and the wavelet spectrum is defined to detect the occurrence of the surface jumps. The approach is shown to be able to produce the same breaking statistics as field measurements of wave breaking conditions based on the detection of whitecaps at fixed points of observation.

Lin and Liu (2004) presented an application of the Wavelet transform for the analysis of freak waves events (in-depth discussion of freak waves is given in Chapter 8). The local wavelet energy density of the wave train is used successfully to determine the groupiness degree of waves and its variation cross-shore (Liu, 2000; Dong *et al.*, 2008), as well as for the separation of wave fields into incident and reflected waves (Ma *et al.*, 2010). Such separation can be performed in the real time domain, also for non-stationary wave trains.

Elsayed (2006a) and Dong *et al.* (2008) demonstrated that the Wavelet transform has become a powerful tool for study of the nonlinear wave-wave interactions of ocean waves. Let us first define the wavelet power spectrum integrated over a finite time interval T

$$E_{xx} = \int_T WT_x^*(\tau, b) WT_x(\tau, b) d\tau, \quad (1.165)$$

where $WT_x(\tau, b)$ is the wavelet transform of a given time series $x(t)$.

In a similar way, the wavelet cross-spectrum takes the form

$$E_{xy} = \int_T WT_x^*(\tau, b) WT_y(\tau, b) d\tau, \quad (1.166)$$

when $WT_y(\tau, b)$ is the wave transform of the second time series.

To the contrast to classical bispectrum (see Section 3.2.4), they defined the wavelet based cross-bispectrum as a triple product of Wavelet transforms as follows

$$B_{yxx}(b_1, b_2) = \int_T WT_y^*(\tau, b) \cdot WT_x(\tau, b_1) \cdot WT_x(\tau, b_2) d\tau, \quad (1.167)$$

where the scale dilation parameters b_1 , b_2 and b must satisfied the frequency sum rule

$$\frac{1}{b} = \frac{1}{b_1} + \frac{1}{b_2}. \quad (1.168)$$

The wavelet based cross-bispectrum is the measure of the phase coupling in the interval T that occurs between wavelet components of time scales b_1 and b_2 of the time series $x(t)$ and wavelet component b of time series $y(t)$,

satisfying Eq. (1.168). In practical applications, the following normalised squared wavelet cross-bicoherence is used

$$[b_{yxx}(b_1, b_2)]^2 = \frac{|B_{yxx}(b_1, b_2)|^2}{[\int_T |WT_x(\tau, b_1)WT_x(\tau, b_2)|^2 d\tau][\int_T |WT_y(\tau, b)|^2 d\tau]}. \quad (1.169)$$

In particular, Elsayed (2006b) showed that phase coupling occurs between wind speed and wave height over a certain range of frequencies during wave growth in a *Mistral* event. However this range is different from one segment to another due to the nonstationarity of the time series.

1.4.4.2 The Hilbert transform representation of wave signal

The wavelet technique, based on Fourier spectral methodology with adjustable frequency-dependent window functions — the mother wavelets — provides temporal-frequency resolution for non-stationary signals. However, the Fourier methodology interprets wave nonlinearity in terms of harmonic generation with the spectral energy leaking into the higher-frequency components. On the other hand, the Hilbert transform interprets wave nonlinearity as frequency modulation, and the spectral energy remains near the base frequencies.¹⁰

To determine the function $\xi(t)$, let us assume for a moment that a real-valued function $\zeta(t)$ extends over the range $-\infty < t < \infty$. Then its Hilbert transform becomes (Bendat and Piersol, 1986)

$$\xi(t) = H[\zeta(t)] = P \int_{-\infty}^{\infty} \frac{\zeta(t_1)}{\pi(t - t_1)} dt_1, \quad (1.170)$$

in which P denotes the Cauchy principle value. Eq. (1.170) indicates that $\xi(t)$ is the convolution integral of $\zeta(t)$ and $1/\pi t$.

If the signal $\zeta(t)$ is represented in the form of a series, we can write

$$\zeta(t) = \sum_{n=0}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)] \quad (1.171)$$

and

$$\xi(t) = \sum_{n=0}^{\infty} [a_n \sin(n\omega t) - b_n \cos(n\omega t)]. \quad (1.172)$$

¹⁰The Hilbert transform is named after David Hilbert (1862–1943), a German mathematician who developed many important tools used in modern mathematical physics. For example, he introduced the extension of the real signal into the complex plane such that it satisfies the Cauchy–Riemann equations (see Eq. (1.15)).

The digital computation of the Hilbert transform was explained by Bendat and Piersol (1986) and Massel (2007), where also was demonstrated that the Hilbert transform of the velocity potential function appears to be the stream function of the water motion (see also Eq. (1.15)). Here for our applications we consider a record of N values ($n = 0, 1, 2, \dots, N-1$) of function $\zeta(t)$ sampled at Δt . Thus the Fourier transform $Z(k\Delta f)$ of $\zeta(t)$ becomes

$$Z(k\Delta f) = \Delta t \sum_{n=0}^{N-1} \zeta(n\Delta t) \exp\left(-i \frac{2\pi kn}{N}\right), \quad (1.173)$$

where $\Delta f = 1/(N\Delta t)$. The values of $Z(k\Delta f)$ are needed only from $k = 0$ up to $k = (N/2)$, where the Nyquist frequency occurs. Now using the values of $Z(k\Delta f)$ we can obtain the $\xi(n\Delta t)$ values as follows

$$\xi(n\Delta t) = 2\Delta f \Im \left[\sum_{k=0}^{N/2} Z(k\Delta f) \exp\left(i \frac{2\pi kn}{N}\right) \right], \quad (1.174)$$

in which $\Im$ denotes the imaginary part of the given value.

Combining the real-valued function $\zeta(t)$ with its Hilbert transform $\xi(t)$ we obtain a complex analytical signal in the time domain

$$\eta(t) = \zeta(t) + i\xi(t) = A(t) \exp(i\theta(t)), \quad (1.175)$$

in which $A(t)$ is interpreted as an instantaneous amplitude

$$A(t) = \sqrt{\zeta^2(t) + \xi^2(t)} \quad (1.176)$$

and $\theta(t)$ is an instantaneous phase

$$\theta(t) = \arctan\left(\frac{\xi(t)}{\zeta(t)}\right). \quad (1.177)$$

From Eq. (1.175) the real part of the analytical signal is the given signal $\zeta(t)$, i.e.

$$\zeta(t) = \Re(\eta(t)) = A(t) \cos(\theta(t)). \quad (1.178)$$

Therefore instantaneous frequency ω can be determined as

$$\omega = \frac{d\theta(t)}{dt}. \quad (1.179)$$

In contrast to the instantaneous envelope of a signal, the notion of an instantaneous frequency is somewhat controversial. When applying Fourier analysis, we need at least one full oscillation to define the local frequency

value. Therefore, a process shorter than a full wave cannot be considered in a signal analysis. However, a straightforward application of the Hilbert transform sometimes leads to the appearance of positive and negative frequency values. To overcome this difficulty, Huang *et al.* (1998) introduced the empirical mode decomposition method. They found that at any given time, there is only one frequency value which can only represent one component. To satisfy this requirement, some limitations on the data are necessary.

Probably the most obvious restriction is the assumption that the process under the consideration is of narrow band, when the bandwidth parameter ν (see Eq. (3.9)) is much smaller than 1, and the expected numbers of extremes and zero crossings have to be equal. However, the bandwidth definition given by Eq. (3.9) has a global sense and it lacks precision. This means that filtering the data within the bandwidth requirement does not necessarily guarantee a meaningful instantaneous frequency. In order to avoid such difficulties Huang *et al.* (1998) suggested applying the Gabor (1946) statement that for any function to have a meaningful instantaneous frequency, the real part of its Fourier transform must have only a positive frequency. Subsequently Huang *et al.* (1998) modified this requirement by imposing the restriction that the function should be symmetric with respect to the local zero mean, and have the same numbers of zero crossings and extremes. To define the local zero mean, the local mean of the envelopes determined by the local symmetry is used. The first restriction is equivalent to the traditional narrow band requirements for a stationary Gaussian process. However, the second requirement in fact modifies the classical global requirement to a local one in a such a way that the instantaneous frequency will not have the unwanted fluctuations induced by asymmetric wave forms.

A modification of the Hilbert transform, is known as the Hilbert–Huang transform (HHT) — see Huang and Shen (2005). Pascoal *et al.* (2005) and Veltcheva (2002) discussed the performance of the HHT and determined an adequate stopping criterion in order to obtain physically meaningful results, and to explain the cross-variation of the group characteristics.

The applications of the Hilbert transform methodology for the wave signals in coastal zone is given in Chapters 9 and 10.